Certifying behaviour rates of language-model policies: what holds, what a label buys, and what does not carry

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Abstract

A developer releasing a language model wants to say something like "this model gives a harmful answer to at most 5% of requests", and to be wrong about it rarely. We call the frequency of an unwanted behaviour (a harmful answer, a needless refusal, a successful attack) its *breach rate*, and a tested statement that the breach rate is below a limit a *certificate*. A Seldonian algorithm (Thomas et al., 2019) returns a trained model only if such a certificate passes on held-out data. We test when the certificate can be trusted, and how to get it from fewer labelled responses, on cases where the true breach rate is known: simulated models, resampled real responses and published benchmark runs.

Standard exact methods can be trusted on data sampled as they assume: promised to be wrong in at most 5% of repeated trials, the standard one was wrong in at most 5.3% of ours, a difference within the noise of the check. Two published shortcuts that let an automatic judge stand in for some labels are guaranteed only for large samples. At the sample sizes of a safety test, in our implementation, they were wrong more often than promised in most of our tests with a judge's score: in up to 24% of trials (a 1.3% rate with 225 labels), and in 90% under an allocation rule published with one of them. A repaired version kept the promise under narrower conditions and ran slightly over on new prompts.

A simpler design makes each label go further. Before training, sample the starting model 8 times on every prompt, sort the prompts into 8 equal groups by how often it breaches, and label the same number of prompts from each group. On new prompts, with predictions registered in advance, a certificate computed from this grouped sample with an approximate formula was wrong no more often than promised at breach rates of 12%, 46% and 82%. The grouped sample did the work of a random sample 1.5 to 1.7 times its size at 12% and 2.1 to 2.6 times at 46%, where much of the gain comes from mixing two prompt sources. At 82% it gained little or lost, at 98% it lost, and for rare behaviours (under 4%) it gained nothing on balance; for rare and near-certain behaviours exact methods remain the ones to use. A sharper formula gave up to 3.0 but ran over its promise at 82%. The gain is smaller, 1.3 to 1.5 at the first two rates, when the certificate must cover all prompts of that kind and not only those sampled. Fifteen of eighteen registered predictions held, two failed and one could not be tested; ten were written after a trial run on older responses. The approximate formula has no guarantee: on simulated data it was wrong too often. These tests use one small model, one benchmark's prompts and an automatic judge's labels, and the design pays only where a label on the trained model costs far more than the 8 judged responses per prompt it needs, a case we did not test.

Three things do not carry over. A judge's measured error rates did not hold on different prompts, nor, in the one training run we examined, on the model after training aimed at the judged behaviour. Methods that treat test cases as independent fail on benchmarks that pair tasks with attacks: on AgentDojo's published runs the usual certificate is wrong too often for 27 of 28 agent configurations, and with the methods that allow for the pairing and kept their promise in our checks, at most one of the 28 certifies an attack success rate under 5%. And a labelling sheet that deliberately over-samples some responses cannot be read as a random sample: combined with a judge's scores it gave certificates that were wrong in up to 99% of trials.

Every certificate of a trained model here rests on an automatic judge's labels. One resting on human labels is designed and costed, not run: if training left the refusal rate unchanged, about 420 labelled prompt pairs would give an even chance of certifying, with an approximate formula, that it rose by no more than 2 points; the exact one is about twice as wide.

1 Introduction

Reports on language-model policies state rates: how often a model refuses a benign request, how often a tool-using agent follows an injected instruction, how often a robot controller trips a safety stop. A decision to deploy needs a different object: a statement that the rate of the unwanted behaviour is at most some threshold, together with the chance that the statement is wrong. We call that statement a certificate.

The Seldonian framework (Thomas et al., 2019) produces this object as the last step of a learning algorithm. The algorithm searches for a solution on one part of the data, tests it on a part the search never saw, and returns it only if a one-sided confidence bound at level delta puts the constrained quantity under its threshold. Otherwise it returns "no solution found" (NSF). Sections 2 to 4 set out the algorithm, its use to post-train a language model, and the theory of each step.

Three features of language-model evaluation can each void the guarantee without any visible sign. The label is not observed: a guard model, a rubric judge or a benchmark harness produces it, and its errors differ from one population of responses to the next. The sample is rarely independent: benchmarks cross tasks with attacks, and failures cluster. And the policy has often been trained against the label that is then used to certify it. The rates of interest are also small, which is where normal approximations are known to fail (Bowyer et al., 2025).

This paper does not propose a new way to train. It takes the safety test, applies it to policies that are fixed by the time they are tested, and asks when its guarantee holds for a language model, what it costs in labels, and what breaks it. Section 5 states the proposal, section 6 the set-up of the evaluation, and sections 7 to 10 the results.

2 The Seldonian algorithm

A standard learning algorithm returns the parameters θ that score best on an objective, and promises nothing about a constraint that entered as a penalty. A Seldonian algorithm (Thomas et al., 2019) starts from the promise. The user supplies constraint functions g_i, with $g_i(\theta) \le 0$ meaning the behaviour is acceptable, and a tolerance δ_i for each. The algorithm a maps data D to a solution or to NSF, and must satisfy

$$\mathbb{P}(g_i(a(D)) \le 0) \ge 1 - \delta_i \qquad \text{for every } i,$$

where $g_i(\text{NSF}) = 0$ by definition, so declining never counts as a violation. With $\delta = \sum_i \delta_i$, the probability of returning a solution that violates some constraint is at most δ. The standard construction has three steps.

```
1. split      D into a candidate set D_c and a safety set D_s
2. candidate  using D_c only, search for the theta_c with the best objective
   selection  among those predicted to pass step 3
3. safety     using D_s only, once: compute U_i, a (1 - delta_i) upper confidence
   test       bound on g_i(theta_c), from unbiased estimates g_hat_i of it
              return theta_c if every U_i <= 0, otherwise return NSF
```

It works because θ_c is a function of D_c alone. Given D_c it is one fixed solution and D_s is a fresh sample, so step 3 is an ordinary confidence bound and falls below the true $g_i(\theta_c)$ with probability at most δ_i. The search may look at any number of candidates, because exactly one is tested, once.

The guarantee is on the output. It does not depend on the search converging or on the prediction in step 2 being accurate. Nothing is promised about how often a solution is returned: an algorithm that always returns NSF is Seldonian, so the rate at which solutions are returned is reported as a measure of quality. And the guarantee is exact only if the bound is. With a normal-approximation bound such as Student's t, Thomas et al. call the algorithm quasi-Seldonian.

3 Seldonian post-training of a language model

Post-training starts from a released, instruction-tuned model and changes its behaviour with a small number of further updates. In the work this paper builds on, the updates are reinforcement learning on a reward (GRPO, Group Relative Policy Optimisation) applied to a LoRA adapter, with the base weights frozen. The untrained base model is the *reference policy*. The concern is that optimising a reward model's score moves behaviours the reward does not measure, such as how often a benign request is refused. Table 1 gives each part of section 2 its language-model form, as implemented in `seldonian/llm/`.

Table 1. The Seldonian algorithm instantiated for post-training a language model.

part of the algorithm	**language-model form**
solution θ	the weights of a LoRA adapter on a frozen instruction-tuned model
data D	prompts, each with a group tag (benign, adversarial, a demographic value). Responses are sampled from the policy when needed
split	60/40 into D_c and D_s, stratified by group, with a fixed seed. D_s is sealed during training and read by one safety test only
objective	the expected reward of the policy's responses, from a reward model or exact match
constraint g_i	the rate of a judged event among responses to a group of prompts, minus a threshold τ_i. The judge is a frozen function of (prompt, response): a guard model's field, or code
threshold τ_i	absolute, or the reference policy's rate plus a margin
candidate selection	GRPO on D_c; the checkpoint that passes a predicted safety test with the highest reward
safety test	one fresh response per safety prompt, judged; a one-sided upper bound on the rate at level δ/k
NSF	the run returns no adapter

Two things differ from the classical setting. The policy can be queried, so $g(\theta)$ needs no off-policy estimate with importance weights: the candidate's own response to each safety prompt is sampled and judged, and the safety test is a bound on a mean of bounded, independent variables. And the constrained quantity is not observed. It is whatever the judge says it is, so every guarantee below is about the judge's label (sections 8.2, 9 and 10.3).

The safety test does not depend on how the candidate was produced. Reinforcement learning, supervised instruction tuning on D_c, or no training at all leave it unchanged. We have run the first (our own adapters) and the last (other people's published models, sections 10.1 and 10.2), and not supervised instruction tuning.

4 Theory: what is optimised, $\hat{g}$, candidate selection, the safety test

4.1 The problem

Write x for a prompt, $y \sim \pi_\theta(\cdot \mid x)$ for a response, $r(x, y)$ for the reward, and $c_i(x, y)$ in $[0, 1]$ for judge i's label, with 1 meaning the constrained event happened. The problem is

$$\begin{aligned} &\text{maximise} && J(\theta) = \mathbb{E}_x \mathbb{E}_y[r(x, y)] \\ &\text{subject to} && g_i(\theta) = \mathbb{E}_{x \in \text{group } i} \mathbb{E}_y[c_i(x, y)] - \tau_i \le 0, \qquad i = 1, \ldots, k. \end{aligned}$$

The first term of g_i is the *behaviour rate* $p_i(\theta)$: the probability that a response to a prompt from group i is flagged, over the prompt and over the policy's own sampling. No algorithm can solve this problem exactly from a sample. The Seldonian requirement replaces it: return a policy with high reward, and only if $g_i(\theta) \le 0$ can be supported at confidence $1 - \delta$. The optimiser increases a surrogate of J on candidate prompts, and a selection rule picks one checkpoint of the run. Neither enforces the constraint. The safety test does.

4.2 $\hat{g}$: the estimate of the constraint

For a prompt x_j drawn from group i and one response y_j sampled from the policy at temperature 1, let $z_j = c_i(x_j, y_j)$. Then

$$\hat{g}_j = z_j - \tau_i, \qquad \mathbb{E}[\hat{g}_j] = p_i(\theta) - \tau_i = g_i(\theta).$$

Each $\hat{g}_j$ is an unbiased estimate of $g_i(\theta)$, and estimates on different prompts are independent when the prompts are an independent sample. This is the $\hat{g}$ of Thomas et al. (2019). Three quantities are built from the labels z_j and should not be confused: the point estimate $\text{mean}(z) - \tau$, used for reporting; the *predicted* upper bound, computed on candidate prompts during training and used to choose a checkpoint (section 4.3); and the *safety* upper bound, computed once on the safety set, which alone decides whether the policy is returned (section 4.4).

Relative thresholds. Several constraints here are relative: the trained policy's rate may exceed the reference model's by at most a margin. In the training runs cited, the reference rate was measured once on candidate prompts and $\tau =$ reference rate $+$ margin was then treated as a constant, so the certificate is about that number and not about the true difference of the two rates. Section 10.4's design does not do that. In human terms the reference rate is unknown, both policies answer the same safety prompts, and the estimate is the paired difference $d_j = z_j(\theta) - z_j(\theta_0)$ in $\{-1, 0, 1\}$.

4.3 Candidate selection

The update. At each step GRPO samples G responses to each prompt of a batch, scores them, and weights a clipped policy-gradient step by the score normalised within the prompt's group of responses, with a KL penalty to the reference policy. The score is centred and scaled within the group, so a penalty acts only where the responses to one prompt differ in it. In the *filter-only* variant GRPO sees the reward r, and the constraint enters through the choice of checkpoint alone. In the *Lagrangian* variant it sees $r - \sum_i \lambda_i c_i$, each penalty applied on its constraint's prompts: a stochastic ascent step on $L(\theta, \lambda) = J(\theta) - \sum_i \lambda_i g_i(\theta)$.

The predicted safety test. Every few steps the current checkpoint is sampled on m candidate prompts, one response each, and judged. The bound the safety test will use is computed on these labels at the sample size n_{eff}, with an optional inflation $\kappa \geq 1$:

$$n_{\text{eff}} = 1/(1/m + 1/n_s),$$
$$U_{\text{pred}} = \text{mean}(z) + \kappa \cdot (U(z; \delta_i, n_{\text{eff}}) - \text{mean}(z)).$$

m and n_s count the candidate and safety prompts in the constraint's group. What has to be anticipated is the gap between the rate seen now on m prompts and the rate the safety test will see on n_s others, and that gap has variance proportional to $1/m + 1/n_s$. Thomas et al. (2019) use the safety-set size with $\kappa = 2$; the later runs cited here use n_{eff} with $\kappa = 1$. A checkpoint is *feasible* if $U_{\text{pred},i} \leq \tau_i$ for every constraint.

The multiplier and the choice. In the Lagrangian variant, after each predicted test,

$$\lambda_i \leftarrow \text{clip}(\lambda_i + \eta \cdot (U_{\text{pred},i} - \tau_i), \text{floor}, \lambda_{\max}).$$

The step size under slack can be set separately, and in the language-model runs cited it was the same. The floor is zero until it is switched on, from the first update or once the constraint has been predicted infeasible. The predicted upper bound drives the multiplier, not the point estimate, so the penalty rises before the estimated rate reaches the threshold. Among feasible checkpoints, the one with the highest mean reward on its predicted-test sample is kept. If none was feasible, the last one is tested anyway and usually fails.

Why none of this carries a guarantee. Lagrangian theory is about a saddle point: for a convex problem, at convergence, the average of the iterates satisfies the constraint in expectation. It says nothing about the checkpoint in hand after 200 steps on a finite sample. The predicted test is computed on candidate prompts, with fresh responses, and the checkpoint is chosen because it looked best on them, so its predicted rate tends to be optimistic (Table 2). Candidate selection decides how often the safety test passes and how much reward the returned policy has. It has no effect on whether the guarantee is valid.

Table 2. "The constraint held": a Lagrangian's statement against a certificate's.

	penalty or Lagrangian	**certificate**
measured on	fresh responses to candidate prompts, which the selection rule has seen	one draw of D_s, independent of both
statement	the estimate ended below the threshold in this run	the (1 - delta) upper bound on the returned policy is below the threshold
selection bias	uncorrected: predicted 0.133 against 0.184 on the safety set (one run of the companion paper); 0.141 against 0.168 (the constrained run of section 9.2). At another seed of the first experiment the predicted test erred the other way, by 0.01–0.02	removed by the fresh draw; the first of those runs returned NSF
checked by	how many runs ended over the threshold: with a fixed penalty of 1, 3 of 3 seeds; with a penalty of 4, 1 of 3	the miss rate against delta (section 7.1)
can decline	no	yes; where training was repeated over seeds the solution rate is reported (section 7.2)

4.4 The safety test

The test. Training has ended and one checkpoint θ_c is selected. For each constraint i, sample one response to each of the n_i safety prompts in its group, judge them, and compute a one-sided upper

bound U_i on the rate $p_i(\theta_c)$ at level $\delta_i = \delta/k$. Return θ_c if $U_i \le \tau_i$ for every i. Otherwise return NSF. The safety set is used once.

The guarantee. A bound is *valid* at level δ_i if $\mathbb{P}(U_i < p_i) \le \delta_i$ for a fixed policy and an independent sample. Given D_c and all the randomness of training, θ_c is fixed and the safety labels are an independent sample with mean $p_i(\theta_c)$. A violating policy is returned only if $U_i \le \tau_i < p_i(\theta_c)$ for some i, so

$$\mathbb{P}(\text{returned and } g_i(\theta_c) > 0 \text{ for some } i \mid D_c) \le \sum_i \mathbb{P}(U_i < p_i(\theta_c) \mid D_c) \le \delta.$$

Averaging over D_c gives the guarantee of section 2, for any selection rule on the candidate side.

The bounds. For 0/1 labels with k positives among n, the Clopper-Pearson bound is exact:

$$U = \text{Beta}^{-1}(1-\delta; k+1, n-k), \qquad \text{with } k = 0 : U = 1 - \delta^{1/n}.$$

For labels in $[0, 1]$ the exact choices are Hoeffding's and Bentkus's bounds and a betting bound (Waudby-Smith and Ramdas, 2024). Student's t is approximate. Section 6.3 gives the bounds for stratified, judge-assisted and clustered data. As a worked case: in the constrained run of section 9.2 the returned checkpoint was step 175 of 200; on the 500 prompts of the safety set its refusal rate by the guard's flag was 0.168, the Clopper-Pearson upper bound at delta 0.05 was 0.198 and the threshold 0.206, so the adapter was returned.

NSF. The test returns NSF when $U_i > \tau_i$ for some constraint. NSF does not say the policy violates. It says this sample cannot support the claim at this confidence, because the rate is too close to the threshold or the sample is too small. Roughly, a policy with true rate p passes with probability one half when $\tau - p = z_{1-\delta}\sqrt{p(1-p)/n}$. With no positive observed, the smallest sample that certifies a threshold τ is $n \ge \ln(\delta)/\ln(1-\tau)$: 299 labels for 1% at delta 0.05, 149 for 2%, 59 for 5%. A method that always returns NSF satisfies the guarantee, so the solution rate is part of the result: 10 of 10 on a brevity constraint, 2 of 3 on over-refusal.

4.5 What the guarantee assumes

The argument of section 4.4 uses four assumptions. Each is routinely violated in language-model evaluation, and each has a part of this paper.

	assumption	**how it fails**	**where we test it**
A1	the safety set had no part in choosing the policy, and is used once	a split or a selection rule that uses safety-set labels; a test repeated until it passes	section 7.2
A2	the safety labels are independent draws from the population the claim is about	benchmarks that cross tasks with attacks; a labelling sheet that over-samples flagged responses	sections 9.3, 9.5, 10.2
A3	the bound is valid at this sample size and this rate	normal and Student-t limits at rates of a few percent	sections 7.1, 8, 9.4
A4	the label is the quantity the claim is about	a judge's errors, corrected by a calibration measured on other responses	sections 9.1, 9.2, 10.3

5 What this paper proposes

Sections 2 to 4 describe an algorithm with two halves: a search for a candidate, and a test that decides whether the candidate may be returned. The search is the subject of the companion paper. This paper is about the test.

The proposal. We propose the Seldonian safety test, taken on its own, as a certificate for a behaviour rate of a language-model policy: a one-sided bound at level delta on the rate of a code- or model-defined label, computed on a sample the choice of policy never saw, with NSF as an outcome. The policy may be one we trained or one somebody else published. The rest of the paper audits the four assumptions of section 4.5. Our contributions:

- **Which bounds hold** their level at the sample sizes and rates of a safety test, with the bounds we use measured under one rule (sections 7 and 8; Table 4 holds the main ones).
- **What a label budget buys:** a safety set stratified by the reference model's own per-prompt rate, and a judge used as a variance reducer under a routing rule that reads only the labels' own counts (section 8).
- **What does not carry:** a judge's calibration, across prompt populations or, in the one run we examined, across training that targets the label; independent-sample bounds on crossed designs; and a stratified labelling sheet read as a random sample (section 9).
- **Certificates on published traces** of a robot benchmark and an agent-security benchmark (sections 10.1 and 10.2).
- **The price of human terms:** a guard model measured against human refusal labels, and the cost of a certificate of a trained policy in those labels (sections 10.3 and 10.4).

What the paper is not. It is not a new training method, and it does not claim that any model is safe. Nor is it a study of NSF for its own sake. NSF is one of the test's two answers. The option to decline is what makes the guarantee possible without a fallback: a method that must return something can promise nothing about it unless a solution known to satisfy the constraint is at hand. How often the test declines is its cost. Here NSF is the answer for 1 of 3 training seeds on the over-refusal constraint, for all three models of a robot benchmark at a 5% threshold and delta 0.05, and for 27 of 28 pipelines of an agent-security benchmark at the same threshold and delta, with its injection tasks taken as fixed (sections 7.2, 10.1 and 10.2). A solution rate is defined over repeated training runs and is reported there; a published trace has one sample and one answer.

Scope.

- The label is a guard model's or the benchmark harness's unless marked human. The human-label results are in section 10.3, with one more in sections 8.2 and 9.5: a harm sheet of 225 responses labelled by one annotator, an author, 3 of them harmful. Section 10.4 costs a certificate in human labels and does not compute one.
- Policies trained here are 0.5B to 2B parameters on one 12 GB card. The frontier-model results are certificates on other people's published traces, not training.
- A certificate is about the label as defined and the population the safety sample was drawn from. It says nothing about other prompts or about the policy after further training.
- How the certified policies were trained is the companion paper's subject. Trained checkpoints are resampled as fixed policies in sections 7.1 and 8.1. How a policy was trained matters to the result in two places only: the policy whose human-terms certificate section 10.4 costs, and the negative result of section 9.2, which exists because training targeted the label. Both are one training run, which also supplies section 8.1's pushed label and the policy of its confirmation pools.

6 Evaluation set-up

The object under evaluation is a *certificate procedure*: a sampling design together with a bound that turns the labelled sample into an upper limit U. We ask whether the procedure is valid, and how wide its limit is.

6.1 Validity: the miss rate

Fix a population with a known true rate p, draw a sample S by the design and compute $U(S)$. The procedure *misses* when $U(S) < p$, and is valid at level delta if

$$\text{miss rate } = \mathbb{P}_S(U(S) < p) \leq \delta.$$

This is the one quantity the guarantee of section 4.4 needs from a bound. It cannot be read off a single real sample, where the truth is unknown, so we repeat the draw R times in a setting where p is known and take $\hat{m}$, the share of the R limits that fall below p. If the true miss rate were exactly delta, $\hat{m}$ would have standard error $\text{se} = \sqrt{\delta(1-\delta)/R}$. One rule is applied to every bound, favoured or not. A cell (one bound, design, population, sample size and delta) is

- *over* its level when $\hat{m} > \delta + 2\text{se}$;
- *unresolved* when $\delta < \hat{m} \leq \delta + 2\text{se}$: the check cannot tell it from delta, which is weaker than showing that the bound holds;
- *at or under* otherwise.

With $R = 4{,}000$ and delta 0.05, a cell is over from a miss rate of 0.057. The three classes say little about how large a miss a check could have overlooked, and the rule is not corrected for the number of cells inspected. So every row of Table 4 also carries two numbers, computed the same way for a bound we use and for one we reject. The first is the largest upper end, over the row's cells, of a 95% Clopper-Pearson interval for the cell's miss rate: the check cannot rule out a miss up to that value. The second is the number of cells still over when the two standard errors are widened by a Bonferroni correction for the cells counted together (one row at one delta or under one scheme).

6.2 Three sources of a known truth

A synthetic environment (the whole pipeline). A contextual bandit stands in for the language model, so that the training and testing code of section 3 runs unchanged and the true rate of any policy is an exact expectation: 20,000 contexts, four actions, a uniform reference policy that violates 13% of the time, and a reward whose pressure parameter makes the better actions the more violating ones. The unit of replication is a whole run (split, training, selection, safety test), and the estimated quantity is the Seldonian failure probability itself, $\mathbb{P}$(returned and truly violating). With a real model one replication is hours of training, so this is where assumption A1 is tested.

Resampling real responses (a plasmode; Franklin et al., 2014). A pool of real responses, each with its label, is treated as the population, and its mean is the truth. A replication draws a safety set from the pool by the design under test and computes the bound. The responses, the judge's scores and their dependence are real; only the sampling is simulated. We use three kinds of pool.

- *Per-prompt pools.* 500 prompts, each with $k = 8$ judged responses of the reference model and further responses of each trained checkpoint, generated by Granite-3.3-2B. A replication draws n_s prompts without replacement, at random or within strata, and one response of the

checkpoint for each. The safety set is then 20–40% of the pool and the truth is the pool's own rate: a finite population, which no bound here corrects for, so every bound is wider than it needs to be and the miss rates come out low. We therefore redraw the same cells with replacement within the same strata, at 40,000 draws a cell, which is the limit of a pool far larger than the safety set.

- *Judge pools.* 500 responses, each with a gold label (the guard's refusal field, or an exact word count) and a rubric judge's score. A replication draws responses with replacement and reveals the gold label on the first n. In the cells at a set rate (20%, 5% and 1.3%) the label is drawn at that rate and the score from the pool's scores for that label, so the truth there is the set rate.
- *Labelling sheets.* A sheet is re-drawn by its real sampling rule, which over-samples the responses a guard flags. The labels in this study are planted on the real responses at a known rate; they are not an annotator's.

Resampling clusters (a crossed design). For AgentDojo each pipeline's observed table of (user task, injection task) outcomes is the population and its overall rate the truth. A replication redraws whole tasks with replacement under one of three schemes: (a) user tasks redrawn, injection tasks fixed; (b) the reverse; (c) both. The scheme states what the certificate treats as sampled, and so what population it speaks about.

6.3 The bounds under test

All are one-sided upper limits at level delta; z is a normal quantile.

Labels alone. The bounds of section 4.4, and the Wilson score bound: the smallest $m > \hat{p}$ with $m - \hat{p} \geq z_{1-\delta}\sqrt{m(1-m)/n}$, and 1 when $\hat{p}$ is 1. It puts the variance at the hypothesised rate and not at the estimate, so its width is not zero when no positive is observed: the limit is then $z^2/(n+z^2)$. It is approximate, and for an i.i.d. sample its miss probability can be computed exactly, by enumeration over the binomial counts (section 7.1).

A stratified sample (`b1w`). With strata $h = 1, \ldots, H$ of population share W_h, n_h labels and observed rate $\hat{p}_h$ in each, the estimate is $\hat{\mu} = \sum_h W_h \hat{p}_h$. The stratified Wilson-type bound is the smallest $m > \hat{\mu}$ with

$$m - \hat{\mu} \geq z_{1-\delta}\sqrt{\sum_h W_h^2 p_h(m)(1-p_h(m))/n_h},$$

where $p_h(m)$ shifts every stratum's rate by $m - \hat{\mu}$, clipped to $[0, 1]$. With one stratum it is the Wilson bound. It is a heuristic and not the inversion of a stratified score test: a common shift of every stratum is not the constrained estimate of the rates, and once a rate is clipped the shifted rates no longer average to m. The point $m = \hat{\mu}$ is excluded because it satisfies the inequality with both sides zero whenever every stratum is all zeros or all ones. It is approximate.

The stratified Wald-t limit, its companion in sections 7 and 8, is $\hat{\mu} + t\sqrt{\sum_h W_h^2 p \sim_h (1 - p \sim_h)/(n_h - 1)}$, with $p \sim_h = (s_h + 1/2)/(n_h + 1)$ for s_h positives and a Student-t quantile at Satterthwaite's degrees of freedom. When the claim is about the population a pool of N prompts was drawn from, both limits add $\sum_h W_h(\hat{p}_h - \hat{\mu})^2/N$ to the variance, the part of the pool's own sampling error that lies between strata.

Labels and a judge (PPI++; Angelopoulos et al., 2023b). With n gold labels y, and the

judge's feature f on the same responses and on N_u unlabelled ones,

$$\hat{p} = \text{mean}(y) + \lambda(\text{mean}_{\text{unl}}(f) - \text{mean}_{\text{lab}}(f)),$$
$$\lambda = \text{cov}(y, f)/((1 + n/N_u)\,\text{var}(f)),$$
$$V = \text{var}(y - \lambda f)/n + \lambda^2\,\text{var}_{\text{unl}}(f)/N_u.$$

$\hat{p}$ is unbiased for the gold rate for any fixed λ, because the judge's two means cancel in expectation. The published limit is $\hat{p} + z_{1-\delta}\sqrt{V}$. StratPPI (Fisch et al., 2024) applies the same correction within each stratum and combines the strata with weights W_h, again with a normal limit. PPBoot (Zrnic, 2024) replaces the normal limit by a percentile of bootstrap estimates.

The bootstrap-t limit. For any of these estimators, resample the labelled responses with replacement B times (within strata, for a stratified estimator), redraw each unlabelled mean from its sampling distribution, and recompute the estimate p_b^* and its variance V_b^*. With $t_b = (p_b^* - \hat{p})/\sqrt{V_b^*}$ and q the delta-quantile of the t_b,

$$U = \hat{p} - q\sqrt{V}.$$

It replaces the symmetric normal quantile by the estimated distribution of the studentised error, which is skewed when the rate is small. It is approximate.

Clustered data. With m clusters (user tasks) of sizes n_c and success counts s_c, $\hat{p} = \sum_c s_c / \sum_c n_c$ and

$$\text{se} = \sqrt{m/(m-1) \cdot \sum_c (s_c - \hat{p} n_c)^2} / \sum_c n_c.$$

The limit is the bootstrap-t limit with whole clusters resampled. The naive alternative treats the episodes as independent and applies Clopper-Pearson.

When clusters cross (user tasks by injection tasks), two more bounds are tested. With V_u and V_i the squares of the standard error above for the two kinds of cluster and $V_p = \hat{p}(1 - \hat{p})/(n - 1)$ the same with every pair its own cluster, the multiway variance (Cameron, Gelbach and Miller, 2011) is

$$V_2 = V_u + V_i - V_p,$$

which we replace by the larger of V_u and V_i if it is not positive. One bound is $\hat{p} + t\sqrt{V_2}$, where for t we take a Student-t quantile on one fewer degrees of freedom than the smaller number of clusters. The other is the bootstrap-t limit with user tasks and injection tasks resampled independently (the pigeonhole bootstrap; Owen, 2007) and V_2 as the variance in every resample. Both return 1 when the table holds no success.

A paired difference. For a relative constraint in human terms the data are the differences d_j of section 4.2. The exact limit is the betting bound on $(d_j + 1)/2$, mapped back. The approximate ones are the bootstrap-t limit on mean(d), and PPI++ with the difference of the guard's two logits as the feature.

6.4 What a design or a judge is worth: effective sample size

Width is compared at equal validity, and where a baseline is itself over its level we say so. For a design or estimator A and a baseline R (a simple random sample with a pooled bound: the pooled Wilson bound for the stratified limits) at the same number of labels and the same delta, with w the mean distance from the estimate to the limit,

$$\text{ESS} = (w_R/w_A)^2.$$

An ESS of 2 means the design does with n labels about what the baseline does with $2n$. In the comparisons with StratPPI (Table A1 and the medians of sections 8.1 and 8.2) and in the prompt-source and confirmation gains of section 8.1, w is measured from the truth to the limit, and in section 8.2 the baseline is the labels alone.

Reference-rate strata. One response per prompt has variance $p(1-p)$, part of it between prompts (prompts differ in their rate) and part within (the policy's own sampling). The between-prompt share is the intraclass correlation, ICC. A sample drawn within strata, in proportion to their size, loses the variance that lies between strata; if that is a share G of the total, $\text{ESS} = 1/(1-G)$. A relative constraint already calls for the reference model's responses, so the strata come from them: sample the reference k times per prompt, rank the prompts by the share flagged, and cut the ranking into H equal strata with ties broken at random. This happens before training and uses no response of the trained policy. The resampling studies of section 8.1 hold the candidate fixed, so they do not test a candidate trained on the pool's other prompts; the in-loop check of section 7.2 does, on a synthetic environment. G follows a formula:

$$G \approx \text{ICC}_{\text{cand}} \cdot \rho^2 \cdot \text{rel}(k) \cdot c_H, \qquad \text{rel}(k) = k\text{ICC}_{\text{ref}}/(1 + (k-1)\text{ICC}_{\text{ref}}),$$

where ICC_{cand} is the trained policy's intraclass correlation, ρ the correlation between a prompt's reference rate and its rate under the trained policy, $\text{rel}(k)$ the reliability of a k-sample reference rate, and c_H the share a cut into H strata keeps. ICC_{cand} and ρ belong to the trained policy. Before one of its responses is labelled they have to be assumed (our tool takes $\text{ICC}_{\text{cand}} = \text{ICC}_{\text{ref}}$ and ρ of 1 or 0.8); section 8.1 reports the formula with the two quantities measured and with $\rho = 1$. When the pool of N prompts is itself a sample of the population the claim is about, the gain is capped at about $1/(1 - G + Gn_s/N)$; section 8.1 measures that case.

A judge. With ρ^2 the squared correlation between the gold label and the judge's feature, the variance V of section 6.3 gives $\text{ESS} = 1/(1 - \rho^2 N_u/(N_u + n))$, which tends to $1/(1-\rho^2)$ when unlabelled responses are plentiful. A judge that predicts nothing has λ near 0 and an ESS of 1.

6.5 Carrying a calibration

A cheaper route uses no gold label on the certified policy at all. If the judge flags a true positive with probability s (recall) and a true negative with probability a (the false-alarm rate), its flag rate is $q = sp + a(1-p)$, so $p = (q-a)/(s-a)$. A *carried* bound measures s and a on one population of responses, measures q on the population to be certified, and bounds p from one-sided limits on the three at $\delta/3$ each. It is valid only if s and a are the same on both populations, which is assumption A4 in its sharpest form. We measure the miss rate of the carried bound across two kinds of shift: between prompt populations, and between a reference model and a policy trained from it.

6.6 Models, labels and data

Table 3. The settings of the evaluation.

part	policy	label	sample	truth	delta
whole pipeline (7.2)	synthetic bandit	simulated judge	200 to 5,000 prompts	exact	0.10
labels alone (7.1)	three trained Qwen2.5-0.5B policies	guard's harm field, rate 0.034	200 to 2,400 of 3,600 responses	pool mean	0.10
reference-rate strata (8.1)	Granite-3.3-2B, reference and trained	guard's refusal and safety fields	100 to 200 of 500 prompts	pool mean	0.05, 0.10

Table 3. The settings of the evaluation.

part	policy	label	sample	truth	delta
judge routing (8.2)	500 real responses; a harm sheet of 225 from 4,800	gold: guard's refusal field, a word count; on the harm sheet, 225 human harm labels for the one real estimate and labels planted at 1.3%, 5% and 20% for the validity cells; judge: a rubric	50 to 1,000 responses	pool mean; the planted rate for the sheet	0.05
carried calibration (9.1, 9.2)	a reference model and two trained policies	gold: guard's refusal field; judge: the rubric, six wordings	benign and harmful prompt pools	pool mean	0.05
robot benchmark (10.1)	three published frontier models	harness safety-stop flag	120 trials each, 20 on each of 6 tasks	unknown	0.05, 0.10
agent benchmark (10.2)	28 published pipelines	harness `security` flag	629 to 949 task pairs each	table rate	0.05
guard against humans (10.3)	five published model variants; our two policies	human refusal labels	1,250 and 220 responses	none	95% intervals
human-terms design (10.4)	reference and trained Granite-3.3-2B	human strict refusal (not collected)	300 of 490 prompt pairs	synthetic labels	0.05

The benign prompts are XSTest's safe prompts and OR-Bench's (Röttger et al., 2024; Cui et al., 2025). The guard is Qwen3Guard (Zhao et al., 2025): the 4B model, run 4-bit quantised, except in the results cited from the companion paper (its entries in Table 2, the labels-alone row of Table 3 and the language-model seeds of section 7.2), which used the 0.6B model. Replications per cell are between 4,000 and 40,000 in Table 4, between 200 and 2,000 runs in the whole-pipeline studies of section 7.2, and 2,000 in the design check of section 10.4.

7 Results: validity

7.1 Which bounds hold their level

Section 6.1 defines the miss rate and the rule that sorts every cell into over, unresolved, and at or under its level; section 6.2 gives the three settings in which the truth is known. Table 4 applies that rule to the bounds we use. Three procedures built from them are measured where they appear: `b1w` with the term for a sampled pool (section 8.1), the routing rule taken as one procedure (section 8.2) and the paired limits of section 10.4 (Appendix C).

Table 4. Miss rate against delta for the bounds used, with one rule for every row. Cells are counted as over / unresolved / at or under delta (section 6.1's rule). The next column is the largest upper end of a 95% interval for a cell's miss rate, with the cells still over after a Bonferroni correction for the cells counted with it in brackets. "Exact" means valid at every sample size by construction; "approximate" means valid asymptotically and checked by resampling. Draws per cell are 4,000 or 5,000, and 40,000 for the rows redrawn with replacement. The sheets' 4,000 are 20 plantings of the labels by 200 sheets each, counted as independent draws. The rows on the trained-policy harm pool are the companion paper's printed rates. Their draws were lost, but labels drawn with replacement from a 0/1 pool are a binomial count, so each of those miss rates is a finite sum. The pool's count is not recorded and its printed rate allows 121 to 124 of 3,600. Computed that way at 124, the count that fits best, all 25 printed values are within 1.3 Monte Carlo standard errors of the exact ones; at 121 only 14 are within 2. The rows "redrawn with replacement" repeat the reference-rate-strata cells with each stratum's prompts drawn with replacement (section 6.2). AgentDojo schemes: (a) user tasks resampled, injection tasks fixed; (b) the reverse; (c) both. The rows on StratPPI's own allocations count 26 reference-strata cells: one checkpoint at a 95% rate is left out, where the proportional-allocation rows count all 28. The row for the label pushed by the Lagrangian holds that run's own cells, at n_s 200; the same label at n_s 100, drawn for the StratPPI comparison, is in Table A1 (largest miss 0.024 at delta 0.05). The last five rows are the stratified Wald-t limit of section 8.1, and it and `b1w` on a synthetic grid.

bound	**kind**	**setting**	**delta**	**miss, per cell**	**cells over / unresolved / at or under**	**largest 95% upper limit of a cell's miss (cells over after correction)**
Clopper-Pearson	exact (binary labels)	real harm labels, trained-policy pool, rate 0.034, n 200–2,400	0.10	0.067–0.084	0 / 0 / 5	0.092 (0)
Clopper-Pearson	exact	the plasmodes of section 8.2, every cell	0.05	0–0.052	0 / 2 / 40	0.059 (0)
betting mixture	exact (bounded)	same pool as row 1	0.10	0.007–0.016	0 / 0 / 5	0.020 (0)
Bentkus	exact (bounded)	same pool	0.10	0.023–0.035	0 / 0 / 5	0.040 (0)
Hoeffding, Anderson	exact (bounded)	same pool	0.10	0.000	0 / 0 / 5	0.001 (0)
stratified Wilson-type `b1w`	approximate	4 mid-rate labels (9–95%), real Granite-3.3-2B responses, n_s 100–200, 3 checkpoints	0.05 / 0.10	0.017–0.054 / 0.058–0.097	0 / 3 / 21; 0 / 0 / 24	0.061 (0); 0.106 (0)
`b1w`	approximate	label pushed by the Lagrangian, n_s 200	0.05 / 0.10	0.011–0.023 / 0.040–0.064	0 / 0 / 2; 0 / 0 / 2	0.027 (0); 0.071 (0)
`b1w` on a design-weighted sheet	approximate	sheets re-drawn by their real sampling rule	0.05	0–0.060	1 / 1 / 16	0.067 (0)
PPI++ with a bootstrap-t limit	approximate	the plasmodes of section 8.2, every cell and feature	0.05	0–0.054	0 / 13 / 155	0.061 (0)
cluster bootstrap-t, by user task	approximate	AgentDojo, 28 pipelines, scheme (a)	0.05	0.013–0.051	0 / 2 / 26	0.058 (0)

Table 4. Miss rate against delta for the bounds used, with one rule for every row. Cells are counted as over / unresolved / at or under delta (section 6.1's rule). The next column is the largest upper end of a 95% interval for a cell's miss rate, with the cells still over after a Bonferroni correction for the cells counted with it in brackets. "Exact" means valid at every sample size by construction; "approximate" means valid asymptotically and checked by resampling. Draws per cell are 4,000 or 5,000, and 40,000 for the rows redrawn with replacement. The sheets' 4,000 are 20 plantings of the labels by 200 sheets each, counted as independent draws. The rows on the trained-policy harm pool are the companion paper's printed rates. Their draws were lost, but labels drawn with replacement from a 0/1 pool are a binomial count, so each of those miss rates is a finite sum. The pool's count is not recorded and its printed rate allows 121 to 124 of 3,600. Computed that way at 124, the count that fits best, all 25 printed values are within 1.3 Monte Carlo standard errors of the exact ones; at 121 only 14 are within 2. The rows "redrawn with replacement" repeat the reference-rate-strata cells with each stratum's prompts drawn with replacement (section 6.2). AgentDojo schemes: (a) user tasks resampled, injection tasks fixed; (b) the reverse; (c) both. The rows on StratPPI's own allocations count 26 reference-strata cells: one checkpoint at a 95% rate is left out, where the proportional-allocation rows count all 28. The row for the label pushed by the Lagrangian holds that run's own cells, at n_s 200; the same label at n_s 100, drawn for the StratPPI comparison, is in Table A1 (largest miss 0.024 at delta 0.05). The last five rows are the stratified Wald-t limit of section 8.1, and it and `b1w` on a synthetic grid.

bound	kind	setting	delta	miss, per cell	cells over / unresolved / at or under	largest 95% upper limit of a cell's miss (cells over after correction)
StratPPI estimator with a bootstrap-t limit, proportional allocation	approximate	reference-rate strata, 5 mid-rate labels, n_s 100–200; judge-logit strata, n 100–1,000	0.05	0.024–0.045; 0.001–0.056	0 / 0 / 28; 0 / 5 / 23	0.051 (0); 0.064 (0)
the same estimator and limit, redrawn with replacement	approximate	reference-rate strata, 5 mid-rate labels, n_s 100–200	0.05 / 0.10	0.030–0.050 / 0.069–0.100	0 / 0 / 28; 0 / 1 / 27	0.052 (0); 0.103 (0)
Clopper-Pearson, random draws with replacement (the control)	exact	the same 5 labels and sizes	0.05 / 0.10	0.029–0.050 / 0.050–0.097	0 / 0 / 28; 0 / 0 / 28	0.052 (0); 0.100 (0)
pigeonhole bootstrap-t	approximate	AgentDojo, 28 pipelines, schemes (a); (b); (c)	0.05	at most 0.004; 0.037; 0.042	0 / 0 / 28; 0 / 0 / 28; 0 / 0 / 28	0.007 (0); 0.043 (0); 0.048 (0)
Student-t	fails at low rates	same pool as row 1, n 200–2,400	0.10	**0.101–0.184**	3 / 2 / 0	0.195 (3)
`b1w` and the pooled Wilson bound at rare rates	in the stored design, not over at delta 0.05; over in 2 cells at 0.10	harm labels at 1–2%, n_s 100–200, 3 checkpoints	0.05 / 0.10	0–0.048 / **0–0.126**	0 / 0 / 24; 2 / 1 / 21	0.054 (0); 0.136 (2)
`b1w`, redrawn with replacement	over on the two labels at rates of 65% and above	reference-rate strata, 5 mid-rate labels, n_s 100–200	0.05 / 0.10	**0.017–0.062 / 0.056–0.113**	7 / 1 / 20; 6 / 2 / 20	0.065 (7); 0.116 (4)
pooled Wilson bound, random draws with replacement	over in 6 and 10 of the 28 cells	the same 5 labels and sizes	0.05 / 0.10	**0.031–0.069 / 0.069–0.128**	6 / 0 / 22; 10 / 3 / 15	0.071 (6); 0.131 (10)

Table 4. Miss rate against delta for the bounds used, with one rule for every row. Cells are counted as over / unresolved / at or under delta (section 6.1's rule). The next column is the largest upper end of a 95% interval for a cell's miss rate, with the cells still over after a Bonferroni correction for the cells counted with it in brackets. "Exact" means valid at every sample size by construction; "approximate" means valid asymptotically and checked by resampling. Draws per cell are 4,000 or 5,000, and 40,000 for the rows redrawn with replacement. The sheets' 4,000 are 20 plantings of the labels by 200 sheets each, counted as independent draws. The rows on the trained-policy harm pool are the companion paper's printed rates. Their draws were lost, but labels drawn with replacement from a 0/1 pool are a binomial count, so each of those miss rates is a finite sum. The pool's count is not recorded and its printed rate allows 121 to 124 of 3,600. Computed that way at 124, the count that fits best, all 25 printed values are within 1.3 Monte Carlo standard errors of the exact ones; at 121 only 14 are within 2. The rows "redrawn with replacement" repeat the reference-rate-strata cells with each stratum's prompts drawn with replacement (section 6.2). AgentDojo schemes: (a) user tasks resampled, injection tasks fixed; (b) the reverse; (c) both. The rows on StratPPI's own allocations count 26 reference-strata cells: one checkpoint at a 95% rate is left out, where the proportional-allocation rows count all 28. The row for the label pushed by the Lagrangian holds that run's own cells, at n_s 200; the same label at n_s 100, drawn for the StratPPI comparison, is in Table A1 (largest miss 0.024 at delta 0.05). The last five rows are the stratified Wald-t limit of section 8.1, and it and `b1w` on a synthetic grid.

bound	kind	setting	delta	miss, per cell	cells over / unresolved / at or under	largest 95% upper limit of a cell's miss (cells over after correction)
PPI++ with a normal limit	fails in most cells	the plasmodes of section 8.2	0.05	**0.041–0.241**	126 / 29 / 13	0.255 (87)
StratPPI's normal limit, proportional allocation	fails at these sizes	reference-rate strata; judge-logit strata	0.05	**0.024–0.086; 0.054–0.239**	9 / 5 / 14; 26 / 2 / 0	0.094 (7); 0.252 (25)
StratPPI's normal limit, the paper's allocations (oracle; heuristic)	fails	reference-rate strata; judge-logit strata	0.05	**up to 0.110; 0.217; up to 0.219; 0.90**	9 / 3 / 14; 11 / 0 / 15; 28 / 0 / 0; 24 / 1 / 3	0.119 (8); 0.229 (11); 0.232 (25); 0.912 (24)
StratPPI estimator with a bootstrap-t limit, oracle or heuristic allocation	fails	the same cells	0.05	**up to 0.078; 0.174; up to 0.083; 0.90**	6 / 3 / 17; 11 / 1 / 14; 14 / 11 / 3; 14 / 0 / 14	0.086 (5); 0.185 (11); 0.092 (9); 0.910 (14)
PPBoot, percentile limit (basic; power-tuned)	fails at these sizes	judge-logit cells, unstratified	0.05	**up to 0.133; 0.148**	7 / 6 / 1; 12 / 2 / 0	0.144 (5); 0.160 (11)
Clopper-Pearson over pairs	fails on a crossed design	AgentDojo, 28 pipelines, schemes (a); (b); (c)	0.05	**0.042–0.268; 0.060–0.340; 0.195–0.350**	27 / 0 / 1; 28 / 0 / 0; 28 / 0 / 0	0.282 (26); 0.355 (27); 0.365 (28)

Table 4. Miss rate against delta for the bounds used, with one rule for every row. Cells are counted as over / unresolved / at or under delta (section 6.1's rule). The next column is the largest upper end of a 95% interval for a cell's miss rate, with the cells still over after a Bonferroni correction for the cells counted with it in brackets. "Exact" means valid at every sample size by construction; "approximate" means valid asymptotically and checked by resampling. Draws per cell are 4,000 or 5,000, and 40,000 for the rows redrawn with replacement. The sheets' 4,000 are 20 plantings of the labels by 200 sheets each, counted as independent draws. The rows on the trained-policy harm pool are the companion paper's printed rates. Their draws were lost, but labels drawn with replacement from a 0/1 pool are a binomial count, so each of those miss rates is a finite sum. The pool's count is not recorded and its printed rate allows 121 to 124 of 3,600. Computed that way at 124, the count that fits best, all 25 printed values are within 1.3 Monte Carlo standard errors of the exact ones; at 121 only 14 are within 2. The rows "redrawn with replacement" repeat the reference-rate-strata cells with each stratum's prompts drawn with replacement (section 6.2). AgentDojo schemes: (a) user tasks resampled, injection tasks fixed; (b) the reverse; (c) both. The rows on StratPPI's own allocations count 26 reference-strata cells: one checkpoint at a 95% rate is left out, where the proportional-allocation rows count all 28. The row for the label pushed by the Lagrangian holds that run's own cells, at n_s 200; the same label at n_s 100, drawn for the StratPPI comparison, is in Table A1 (largest miss 0.024 at delta 0.05). The last five rows are the stratified Wald-t limit of section 8.1, and it and `b1w` on a synthetic grid.

bound	kind	setting	delta	miss, per cell	cells over / unresolved / at or under	largest 95% upper limit of a cell's miss (cells over after correction)
cluster bootstrap-t by user task, when injection tasks are sampled	fails	AgentDojo, schemes (b); (c)	0.05	**up to 0.258; 0.275**	21 / 0 / 7; 24 / 0 / 4	0.272 (20); 0.289 (24)
larger of the two clustered bounds	holds under (a) only	AgentDojo, schemes (a); (b); (c)	0.05	at most 0.029; **up to 0.066; 0.089**	0 / 0 / 28; 4 / 0 / 24; 16 / 3 / 9	0.034 (0); 0.074 (3); 0.098 (15)
pigeonhole bootstrap, basic limit	fails	AgentDojo, schemes (a); (b); (c)	0.05	**up to 0.174; 0.178; 0.278**	5 / 0 / 23; 14 / 1 / 13; 27 / 0 / 1	0.187 (5); 0.190 (10); 0.292 (26)
PPI on a stratified sheet read as an i.i.d. sample	fails	three judge wordings, labels planted at three rates	0.05	**0–0.989**	7 / 0 / 11	0.992 (7)
the judge's rate alone	fails	the plasmodes of section 8.2	0.05	**0–1.000**	30 / 0 / 12	1.000 (30)
multiway cluster variance with a t quantile	fails	AgentDojo, schemes (a); (b); (c)	0.05	**up to 0.136; 0.175; 0.179**	5 / 0 / 23; 8 / 3 / 17; 26 / 1 / 1	0.147 (5); 0.187 (7); 0.191 (25)
two clustered margins added in quadrature	over in one cell under (c)	AgentDojo, schemes (a); (b); (c)	0.05	at most 0.015; 0.046; **0.058**	0 / 0 / 28; 0 / 0 / 28; 1 / 1 / 26	0.020 (0); 0.053 (0); 0.066 (0)

Table 4. Miss rate against delta for the bounds used, with one rule for every row. Cells are counted as over / unresolved / at or under delta (section 6.1's rule). The next column is the largest upper end of a 95% interval for a cell's miss rate, with the cells still over after a Bonferroni correction for the cells counted with it in brackets. "Exact" means valid at every sample size by construction; "approximate" means valid asymptotically and checked by resampling. Draws per cell are 4,000 or 5,000, and 40,000 for the rows redrawn with replacement. The sheets' 4,000 are 20 plantings of the labels by 200 sheets each, counted as independent draws. The rows on the trained-policy harm pool are the companion paper's printed rates. Their draws were lost, but labels drawn with replacement from a 0/1 pool are a binomial count, so each of those miss rates is a finite sum. The pool's count is not recorded and its printed rate allows 121 to 124 of 3,600. Computed that way at 124, the count that fits best, all 25 printed values are within 1.3 Monte Carlo standard errors of the exact ones; at 121 only 14 are within 2. The rows "redrawn with replacement" repeat the reference-rate-strata cells with each stratum's prompts drawn with replacement (section 6.2). AgentDojo schemes: (a) user tasks resampled, injection tasks fixed; (b) the reverse; (c) both. The rows on StratPPI's own allocations count 26 reference-strata cells: one checkpoint at a 95% rate is left out, where the proportional-allocation rows count all 28. The row for the label pushed by the Lagrangian holds that run's own cells, at n_s 200; the same label at n_s 100, drawn for the StratPPI comparison, is in Table A1 (largest miss 0.024 at delta 0.05). The last five rows are the stratified Wald-t limit of section 8.1, and it and `b1w` on a synthetic grid.

bound	kind	setting	delta	miss, per cell	cells over / unresolved / at or under	largest 95% upper limit of a cell's miss (cells over after correction)
stratified Wald-t limit, 8 reference-rate strata	approximate	5 mid-rate labels, real Granite-3.3-2B responses, n_s 100–200, the draws of the StratPPI row	0.05 / 0.10	0.007–0.036 / 0.027–0.080	0 / 0 / 28; 0 / 0 / 28	0.042 (0); 0.088 (0)
the same limit, redrawn with replacement	approximate	the same 5 labels and sizes	0.05 / 0.10	0.007–0.038 / 0.026–0.086	0 / 0 / 28; 0 / 0 / 28	0.040 (0); 0.088 (0)
the same limit with the term for a sampled pool	approximate	a claim about the prompt source: the pool redrawn and the strata rebuilt, the same 5 labels and sizes	0.05 / 0.10	0.007–0.050 / 0.031–0.095	0 / 1 / 27; 0 / 0 / 28	0.055 (0); 0.101 (0)
stratified Wald-t limit on a synthetic i.i.d. grid	over in 5 of 18 cells at each delta	six strata profiles, n_s 100–400, binomial strata	0.05 / 0.10	**0–0.111 / 0.086–0.147**	5 / 5 / 8; 5 / 5 / 8	0.121 (4); 0.158 (4)
`b1w` on the same grid	over in 1 and 4 of 18 cells	the same profiles and sizes	0.05 / 0.10	**0–0.057 / 0.086–0.135**	1 / 2 / 15; 4 / 5 / 9	0.065 (0); 0.146 (2)

In the rows we call valid no cell stays over its level after the correction, and the largest miss rate a cell's interval leaves open is 0.067 at delta 0.05 and 0.106 at 0.10. One cell there is over under the uncorrected rule: `b1w` on a design-weighted sheet, 0.0595 at 4,000 draws. Those 18 cells are 9 settings run twice: the other run of the same setting gave 0.0485, and the two together give 0.054, which is unresolved. Every row in bold has at least one cell over, and every one but the quadrature bound keeps one after the correction. Most also have cells that are not over: a failing bound fails somewhere, not everywhere.

Three readings of Table 4.

1. **The Wilson-type bounds are over their level at some rates, and we can say which.** In the stored design neither `b1w` nor the pooled Wilson bound is over in any of the 24 rare-label cells at delta 0.05; at delta 0.10 two are, both at 0.126, for one harm label at the checkpoint where its rate is 1.9%, at n_s 100. An exact calculation and a redraw with replacement, below, give the pattern: under one half a miss below delta on average and above it at some rates, above one half a miss above delta on average, and a peak just above the limit the bound returns when it sees no positive. Rare labels take Clopper-Pearson or a betting bound.

2. **The t-test is the wrong default.** On the one trained-policy pool we resampled (a harm rate of 3.4%) it is over its level at sample sizes up to 800 and unresolved above them, and its width is zero at a rate of 0 or 1, which lets candidate selection win by driving a subgroup rate to the boundary. A policy with a true rate of 1% passes a 0.5% threshold whenever 100 samples hold no positive, which is 37% of the time. The failure is not particular to that pool. By enumeration over binomial counts, the Student-t limit's miss probability is above delta at 96% or more of 79 rates from 0.5% to 20%, at each of five sample sizes from 200 to 2,400 and at both deltas, and Clopper-Pearson's is above it at none.

3. **"Approximate" has a number attached, and it depends on the design.** In the designs they were run in, the approximate bounds we use are over their level in 1 of 378 cells and unresolved in 24; the largest miss at delta 0.05 is 0.060. The 378 are the cells of the seven approximate rows of Table 4 that are not in bold, not redrawn with replacement and not for the prompt source, leaving out the pigeonhole bootstrap-t, which no certificate in this paper uses. The grouping was made after the results. They are not 378 separate studies: the delta 0.10 cells of the two `b1w` rows and of the Wald-t row reuse the draws of their delta 0.05 cells, the 168 PPI++ cells are 42 sets of draws read through four judge features, and the reference-rate cells of the StratPPI and Wald-t rows share their seeds with the `b1w` rows. Two things qualify the count for `b1w`. It is for one setting of the strata, 8 reference samples and 8 strata; at the other eleven settings in the same file `b1w` is over in 8 of 528 mid-rate cells, all at n_s 100 on the two labels at 65% and above. And it is for a safety set that is 20–40% of its pool; redrawn with replacement `b1w` is over in 7 of 28 cells, by at most 1.2 points at delta 0.05, where the StratPPI estimator with a bootstrap-t limit and a stratified Wald-t limit are over in none. On a synthetic i.i.d. grid (36 cells, 4,000 draws) `b1w` is over in 5 cells and the Wald-t limit in 10. Inside the training loop on a synthetic environment the stratified `b1w` test reached 0.116 at delta 0.1 on 500 runs, which is unresolved, and the Wald-t limit 0.132, which is over (section 7.2). On real responses with 8 strata the Wald-t limit was over in no mid-rate cell and, with the sampled-pool term, in one rare-label cell (section 8.1); on synthetic data it is over in 10 of 36 cells.

The Wilson limit, exactly. For an i.i.d. sample the Wilson limit's miss probability can be computed by enumeration. It jumps just above each value the limit can take and falls until the next. Just above the zero-count limit (2.6% at n 100 and 1.3% at n 200, at delta 0.05) it is 0.069 and 0.068, close to exp $(-z^2)$ at any n: 0.067 at delta 0.05 and 0.194 at delta 0.10, where the zero-count limit at n 100 is 1.6%. Averaged over rates, the miss is under delta below one half and over it above. At n 100 and delta 0.05 it averages 0.042 over rates of 5–20% and 0.056 over 80–95%, with largest values of 0.063 and 0.085, and just under a rate of 1 it reaches 0.20. Clopper-Pearson's miss probability never exceeds delta.

Redrawn with replacement. The stored cells hide that pattern, because they draw 20–40% of a finite pool (section 6.2). With replacement, `b1w` is over its level in 7 of 28 mid-rate cells at delta

0.05 (largest miss 0.062), all on the two labels at rates of 65% and above, and in none of the 16 cells at rates of 9–18% (largest 0.034). The pooled Wilson bound is over in 6 of 28 (largest 0.069), two of them at rates of 15–18%. Each of the two is over in 1 of the 12 rare-label cells at delta 0.05, the same harm label at a checkpoint where its rate is 1.4%, at n_s 200 (0.055 and 0.064), and in 3 of 12 at delta 0.10 (up to 0.138 and 0.142). Clopper-Pearson, exact on those draws, is over or unresolved in none, and on the random draw the simulated misses of both bounds agree with the binomial calculation.

The StratPPI rows (Fisch et al., 2024) are our implementation of its estimator and interval on the same pools as the rows above them, and for the reference-rate strata on the same draws; sections 8.1 and 8.2 give the comparison.

7.2 Validity under the full pipeline

Table 4 tests bounds on a fixed population. The certificate also has to survive selection: the candidate is the best-looking checkpoint of a training run.

Synthetic environment, exact truth. In a contextual bandit where every policy's true rate is computable, the probability of returning a policy that truly violates was 0.000–0.002 against a delta of 0.1 for the Lagrangian variant of the Seldonian arm under four bounds, against 0.830 for unconstrained training at the same reward pressure. In two one-way sweeps of that variant it stayed at or below 0.002, across reward pressures 0 to 4 at 1,000 prompts and across sample sizes 200 to 5,000 at pressure 1 (the second with the Student-t bound; the sizes are total prompts, 40% of them in the safety set). The filter-only variant was higher and still under delta: 0.028 at 200 prompts and 0.004 at 1,000 and 5,000. With a noisy judge the guarantee held at the judge's level in every row, while a judge that misses 40% of violations let the true violation rate among returned policies reach 0.059: the certificate is about the label.

Adversarial splits. Rerandomised and stratified candidate/safety splits did not break the exact test, across 2,000 seeds per arm, eleven split rules and candidates built to overfit the balancing statistics, on a classifier with a constraint on a difference of true-positive rates: with Clopper-Pearson a candidate that passed and truly violated was returned in at most 1 of 2,000 runs in any cell. That test is too conservative there to show a leak, so the sensitive check is a tight Wald test on the same runs, whose bound fell below the truth in at most 0.055 of runs at delta 0.05; it was unresolved above delta in 3 of 42 cells.

The training loop, on a synthetic environment. With our `SeldonianLLMPolicy` class training on a synthetic bandit at four levels of between-prompt heterogeneity, with 4 strata and 500 runs per cell, the stratified `b1w` test, with its added term for a pool that is itself a sample and the population's rate as the truth, missed 0.084–0.116 at delta 0.1 (Monte Carlo standard error 0.013): three cells at or under delta and one unresolved above it. The stratified Wald-t limit missed 0.090–0.132 in the same runs, over its level in one of the four cells (0.132, against a band that ends at 0.127). A random split with a pooled bound missed 0.090–0.100, which 500 runs cannot tell apart from the stratified test. No language model is in this check.

Language-model policies. On an over-refusal constraint (Qwen2.5-0.5B, three seeds), unconstrained training and a fixed penalty of 1 ended over the threshold in 3 of 3 seeds, a penalty of 4 in 1 of 3; the Seldonian arm ended over it in 0 of 3 and certified 2 of 3. On a brevity constraint, 10 of 10 seeds certified and none ended over its threshold. Three and ten seeds cannot estimate a miss rate; they are consistent with the level, and the resampling studies above are the evidence.

Trajectories. On a synthetic bandit, a test of the returned policy alone misses drift through a forbidden region during training: 65% of plain-Lagrangian runs passed it after a step inside the region (50% when checked every 10 steps). A certificate at level delta/T over every one of T checks

was at or under delta in all 10 cells, with misses of 0.025–0.100 against a delta of 0.1, at 200 runs a cell, which resolve only a miss above 0.14.

8 Results: what a label budget buys

8.1 Making the sample go further: reference-rate strata

Idea. Section 6.4 gives the design and why it can help: sample the reference k times per prompt, cut the prompts into H equal strata of their reference rate, sample the safety set within strata, and bound the rate with the stratified Wilson-type bound `b1w`. The gain depends on how much of the per-response variance sits between prompts (the intraclass correlation of the reference, $\mathrm{ICC}_{\mathrm{ref}}$) and on how well a prompt's reference rate predicts its rate under the trained policy.

Result. With k = 8 and H = 8 on real Granite-3.3-2B responses, the effective safety-sample size (ESS: the size of a simple random sample giving the same bound width) against a random split at delta 0.10 was 2.4 for over-refusal (rate 17%, $\mathrm{ICC}_{\mathrm{ref}}$ 0.72), 5.1–5.3 for refusal of plainly harmful requests (66%, 0.86), and 1.4 for non-refusal of encoded requests (9%, 0.50), with coverage as in Table 4. At delta 0.05 the three are 2.30–2.35, 4.79–5.14 and 1.39–1.41. A placebo covariate, run with 4 strata, gave 1.00 to 1.04 on every label but the one at 93% (0.83 to 1.02). These are gains of `b1w` for the pool's own rate, and the 66% label is one where `b1w` is over its level once the pool is large (below). When the Lagrangian pushed the stratified label itself, the gain was 2.1–2.2 at n_s 200 and delta 0.10 (2.05–2.16 at 0.05), with no compression of the between-prompt spread (correlation between a prompt's reference and trained rates 0.92), in one run whose rate ended within 0.4 to 3.4 points of the reference's; a push that moves the rate far was not seen.

The rule was chosen on these pools. The strata rule above is not the one we registered in advance (Appendix D). That rule (ties kept together, cuts at quantiles, small strata merged) collapsed to one stratum on three labels and to two or three on the others, because 73–97% of prompts have an 8-sample reference rate of exactly 0 or 1. On the labels at 9% and 17% it gave less than the rule above (1.2 to 2.3 against 1.4 to 2.5 at delta 0.05), and on the 66% label `b1w` was over its level under it in 2 of 6 cells at delta 0.05 (0.073 and 0.092) and 4 of 6 at 0.10. Equal rank strata with random ties were compared with it on the same pools and adopted, and the truth used for coverage was changed to the mean of the half of the responses the draws come from. Under the registered truth, which carried a fixed offset shared by every draw, the stratified arm and the random-split baseline were both over their level on over-refusal with 4 strata (largest miss over checkpoints 0.20 for the stratified arm, 0.14 and 0.18 for the baseline, at delta 0.10). The gains above are therefore measured on the data that chose the rule; the confirmation at the end of this section checks them on pools that played no part in the choice.

A claim about the prompt source. The gains above are for the rate over the pool's own 500 prompts. If the claim is about the source the prompts were drawn from, the pool is a sample as well and the gain is capped (section 6.4). We measured that case on the same responses. Each replication draws a pool of the same size from the stored one, rebuilds the strata on it and draws the safety set from it; the truth is the stored pool's rate, and a cell has 10,000 replications. With the term for a sampled pool (section 6.3), `b1w` is over its level in none of the 28 mid-rate cells at delta 0.05 (4 unresolved, largest miss 0.052) and in one at 0.10 (0.112, on the 93% label); the Wald-t limit is over in none at either. On the two rare labels `b1w` is over in 1 of 12 cells at delta 0.05 and 3 at 0.10, with or without the term, as in section 7.1. The gain at delta 0.05, for safety sets of 100 and 200, is 1.83 and 1.54 for over-refusal (1.68 and 1.44 when the Lagrangian pushes the label), 2.36 and 1.75 for refusal of harmful requests, and 1.35 and 1.25 for non-refusal of encoded requests. For the pool's own rate the same measure (the truth to the limit, the median over checkpoints) gives

2.39 and 2.47, 2.11 and 2.12, 4.66 and 5.14, and 1.50 and 1.52. At the 93% rate (refusal of encoded requests, 93% to 95% by checkpoint) the strata lose (0.88 and 0.93), and at rare rates they gain nothing (1.01 to 1.05). The baseline, the pooled Wilson bound on a random sample, is itself over its level in 7 of these 28 cells, which understates the gains where it is. Section 6.4's cap, with G taken from the measured large-pool gain, gives 1.84 and 1.54, 1.71 and 1.46, 2.68 and 1.92, and 1.35 and 1.24: the cap is met at the lower rates and missed by about a tenth at 66%. Two limits that serve for the pool's rate do not serve here. At delta 0.05 `b1w` without the term is over in 17 of the 28 cells (misses up to 0.170). So is the bootstrap-t StratPPI limit, in 21 (up to 0.166): its variance has a term for the unlabelled sample inside fixed strata and none for strata that are rebuilt on each pool. On the confirmation pools at the end of this section `b1w` with the term is itself over at an 82% rate, and the Wald-t limit with the term is not: for the source's rate at a rate above one half, the Wald-t limit is the one the evidence supports.

Where it does not help. Rare labels (the gain is 1.01–1.09 at rates of 1–2%, and on a synthetic grid exact stratified bounds beat their own pooled form in one profile only); labels near 0 or 1 (at a 93% rate 0.96 and 1.12 for the pool's own rate, Table A1, and 0.88 and 0.93 for the prompt source); a claim about the population the pool was drawn from when the safety set is a large share of the pool (the gain is then capped at $1/(1 - G + Gn_s/N)$); and task strata on a benchmark with 20 trials a task, where the bound is set by the positives.

A pre-flight. The gain can be ranked from the k reference samples and the two quantities of the trained policy in section 6.4's formula, measured on responses the draws do not use: Spearman 0.84 between predicted and realised ESS over 432 cells on real data at delta 0.10, and 0.83 at 0.05 (Figure 1). Its size is over-predicted. At H = 8 the prediction is 2% to 20% high on the three labels at 9% to 66% at the last checkpoint and delta 0.10, up to 30% high over all checkpoints and both deltas, and 18% to 98% high on the 93% label and the rare ones. Before any trained-policy response is labelled those two quantities have to be assumed; with $\rho = 1$ the prediction ran well above the realised gain on new prompts (the confirmation below). We had registered two criteria: a median absolute error of at most 0.1 and a Spearman of at least 0.8. The first failed on real data (0.29) and passed on the bandit (0.02). The second holds under the strata rule adopted afterwards and fails under the registered one (0.71). We report the ranking as the usable part.

Against StratPPI. StratPPI (Fisch et al., 2024; section 6.3) is the published method closest to this one. We ran our implementation of it on the same strata and the same 5,000 draws per cell, with the reference rate as the predictor (Figure 2, left; Table A1 in Appendix A). Four things follow.

1. **StratPPI's bound is shorter, and part of that is the normal limit running over its level.** It is over its level in 4 of 10 cells at delta 0.05 when a cell is the largest miss over checkpoints (one of them marginally, at 0.056), and in 9 of 28 cells counted by checkpoint, most at the 9% label.

2. **With an honest limit the two are close.** With labels allocated in proportion to stratum size, the same estimator with a bootstrap-t limit is over its level in none of the ten cells (largest miss 0.045), and its median ESS is 2.10 against 2.16 for `b1w`. The repair depends on the allocation (Appendix A).

3. **At a safety set of 100 with 8 strata the gain is the stratification.** The within-stratum regression on the reference rate adds nothing there once the limit is valid (median ESS 1.77 against 2.19 for `b1w` over the 13 cells of the sweep). At a safety set of 200 or with 4 strata it adds 0.4 to 0.8 of ESS: the bootstrap-t StratPPI is the narrower, with a median ESS of 2.6–2.7 against 1.8–2.2 for `b1w` over the 13 cells of a sweep by checkpoint (Appendix A).

4. **Which bound to use depends on the pool and the rate.** For the pool's own rate, with a

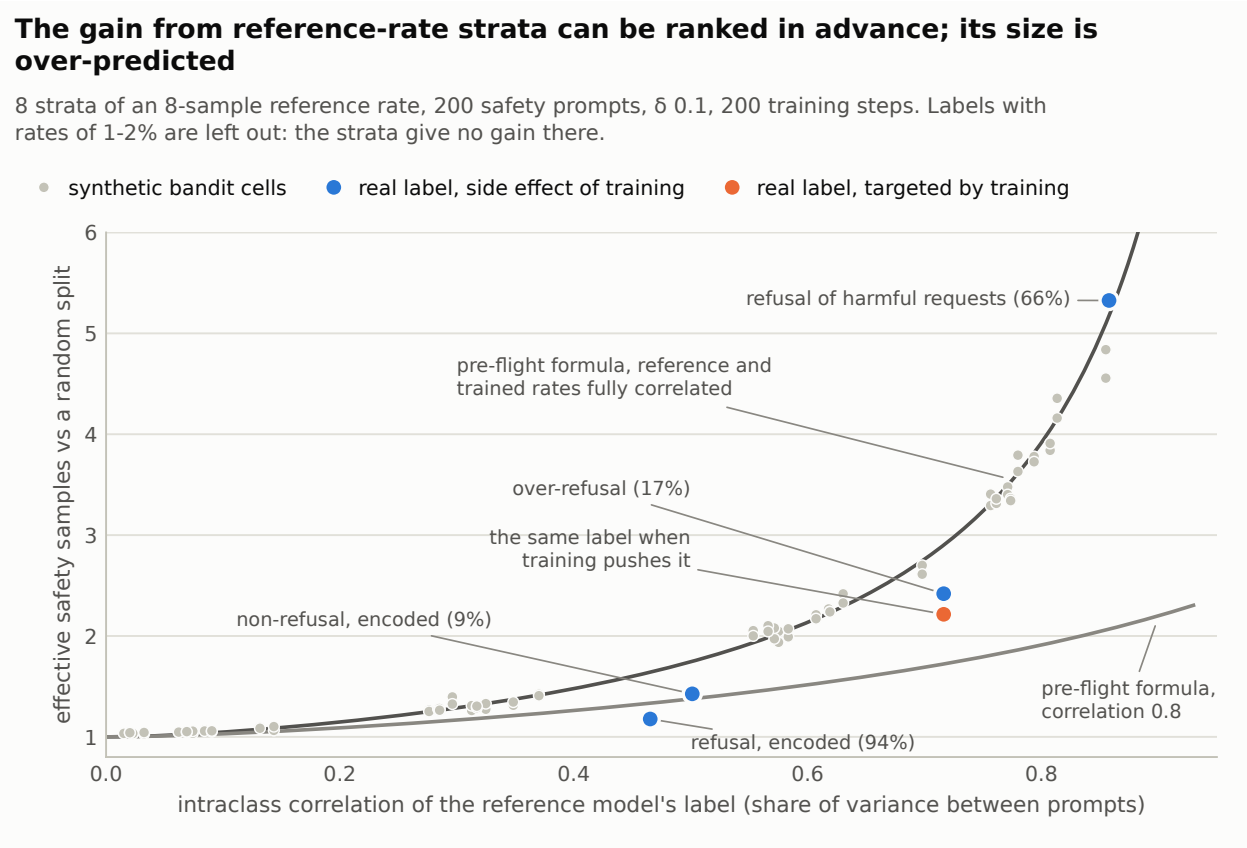


Figure 1. Gain from reference-rate strata against the reference model's intraclass correlation.



Figure 2. StratPPI's normal limit and the same estimator with a bootstrap-t limit, with labels allocated in proportion to stratum size. Left: reference-rate strata. Right: strata on a judge's logit.

safety set of 20–40% of the pool, `b1w` is over its level in none of the 28 mid-rate cells at either delta (3 unresolved at delta 0.05, on the 93% label) and in 1 of its 12 rare-label cells at delta 0.10 (0.126). Redrawn with replacement it is over in 7 of 28 at delta 0.05 and 6 at 0.10, all at rates of 65% and above, and the bootstrap-t StratPPI limit and the stratified Wald-t limit are over in none (section 7.1); the median ESS of the three is 2.16, 1.92 and 1.99. The gains carry over: at n_s 100, `b1w`'s ESS with replacement and without is 2.29 and 2.39 for over-refusal, and 4.73 and 4.74 for refusal of harmful requests. So for a pool much larger than the safety set, `b1w` kept its level at rates of 9–18% and not at 65% and above, which is the asymmetry of the Wilson limit. No label between those rates was tested on these pools, and there the other two limits are what the evidence supports. On the 65% label, where `b1w`'s gain is largest and its level is not kept, they give 2.6 and 3.6 (Wald-t) and 1.55 and 5.47 (bootstrap-t StratPPI) at n_s 100 and 200. Like the strata rule, this is read off the cells it describes; the confirmation at the end of this section bears it out for `b1w` and qualifies it for the bootstrap-t limit.

What the gain costs. The strata are built from k = 8 judged reference responses on each of the pool's 500 prompts, 4,000 in all, and the ESS counts only the safety set's labels. An ESS of 2.4 is worth 140 labels at n_s 100. So the method saves labels only where the label on the trained policy costs far more than the stratifier's, for instance a human label stratified on a guard's reference rate. We did not test that case, and section 10.3 shows the guard and the annotator disagree on about 40% of the guard's flags. Where both are the same cheap judge, enlarging the safety set is the better use of the budget.

Confirmation on new prompts. Everything above was chosen on the pools that measure it. We therefore registered eight predictions (Appendix D lists them) and tested them on three pools of 400 prompts that no earlier pool, training set or result had used: benign prompts of OR-Bench's

80K set, its hard-1K set and its toxic set. The model, the strata rule, the checks and the scoring code are the ones above; the trained policy is the returned checkpoint (step 175) of section 9.2's constrained run. Its refusal rates came out at 12%, 82% and 98%, so no label fell between 18% and 65%, the range no earlier pool had tested either; two labels from the guard's safety field were rare (3.2% and 1.6%). Six predictions were kept and two refuted.

- *Kept.* With a large pool `b1w` is over its level in none of the 4 cells at the 12% rate and in all 4 at 82% (misses 0.062 at delta 0.05 and 0.111–0.113 at 0.10). Its gain is within 20% of the pre-flight's prediction, 2% to 16% under it: 1.53 and 1.62 at 12%, 1.49 and 1.62 at 82%, for safety sets of 100 and 200. For the prompt source the limits without the sampled-pool term fail again (`b1w` in 3 of 4 cells at delta 0.05, the bootstrap-t StratPPI limit in 4 of 4), and `b1w`'s gain with the term is 1.20 to 1.32, within 15% of the cap. On the rare labels the strata gain nothing (1.01 to 1.11) and Clopper-Pearson is over in no cell; the Wald-t limit with the term is over in one rare-label cell for the source (0.110 at delta 0.10, with a safety set of 200).
- *Refuted.* With a large pool the bootstrap-t StratPPI limit is over in 1 of 8 mid-rate cells (0.105 at delta 0.10 on the 82% label), and in one more at the 98% rate (0.052 at delta 0.05); the stratified Wald-t limit is over in none of the 8 (misses at most 0.039 at delta 0.05 and 0.082 at 0.10). And for the prompt source `b1w` with the term is over at the 82% rate with a safety set of 100 (0.058 at delta 0.05 and 0.117 at 0.10; unresolved at 200 and delta 0.05, 0.052), where the Wald-t limit with the term is over in none of 8 (at most 0.047 and 0.088), with a gain of 1.31 and 1.32 at 12% and 0.75 and 0.89 at 82%. Each refuted prediction named the Wald-t limit beside the limit that failed, and its half of each held.

A pool aimed at the gap. No part of OR-Bench that we sampled is both in that range for this policy and large enough to make a pool. By prompt category the three pools' refusal rates run from 0% to 22%, from 46% to 89% and from 87% to 100%; three categories fall inside the range, with 47, 20 and 17 prompts. We therefore registered a fourth pool separately, as a fixed mixture: the 193 hard-1K prompts still unused and 207 unused benign prompts of the 80K set, with ten predictions (Appendix D). It is weaker evidence than the three pools above, in three ways. Its rate was aimed at, not found. Its predictions were written after the same analysis had been run on a re-mix of the first two pools' responses in the same proportions. That loosened one prediction, added one and set the floor of a third, and as finally worded all nine that could be tested already held on the re-mix. And the reference rate largely sorts the two sources into different strata, so part of the gain is the gain of knowing the source: the variance of a random draw is 1.93 times that of a stratified one with two strata, the two sources, and 2.90 times with the 8 strata of the reference rate; by variance removed, the source alone gives about three quarters of what the 8 strata give.

The refusal rate came out at 46% (81% on the hard prompts, 12% on the others). Nine predictions were kept, none was refuted, and one had no label to test it. With a large pool the Wald-t limit is over its level in none of its 4 cells (misses 0.026 and 0.036 at delta 0.05), the bootstrap-t StratPPI limit in none (three cells unresolved, the largest miss 0.001 above its level), and `b1w` in none (0.044 and 0.045 at delta 0.05, 0.096 at 0.10). On the re-mix, at a rate of 48%, `b1w` sat at its level (0.051 and 0.102, unresolved). The gain of `b1w` is 2.78 and 2.97 against a pre-flight of 2.79, and that of the Wald-t limit 2.09 and 2.56. For the prompt source the Wald-t limit with the sampled-pool term is over in none of 4 cells, with a gain of 1.49 and 1.34, 7% to 9% under its cap; `b1w` with the term is over in none either (one cell unresolved, 0.102 at delta 0.10), and the limits without the term fail in both cells at delta 0.05 (misses 0.079 to 0.120). The baseline of these gains, the pooled Wilson limit, is itself over its level at a safety set of 200 and delta 0.05 (0.053, and 0.057 for the source), so the gains at 200 are if anything understated. The safety-field label was rare (1.9%) and the strata

gained nothing on it (1.00 and 1.08); there `b1w` and the pooled Wilson limit are both over at a safety set of 100 and delta 0.10 (0.140 and 0.142), and Clopper-Pearson is not.

Three things follow. First, the gains are smaller than on the pools that chose the rule. On new prompts from one source a limit that kept its level gained 1.5 to 1.7 at 12% (`b1w` 1.53 and 1.62, the Wald-t limit 1.47 and 1.73; the bootstrap-t StratPPI limit, which also held there, 1.55 and 1.88) and 0.86 and 1.13 at 82% (the Wald-t limit; `b1w`'s 1.49 and 1.62 there come from a limit over its level in all four cells). The same measure gives 1.5 to 5.0 for `b1w` at rates of 9% to 66% on the pools that chose the rule, the upper end at 66%, where `b1w` is also over its level with a large pool; the mixed pool's 2.8 to 3.0 is in large part what two unlike sources in one pool give. At all three mid rates, 12%, 46% and 82%, the pre-flight was within 20%, with the trained policy's two quantities measured on 8 of its responses a prompt that the draws do not use. From the reference samples alone, with $\rho = 1$, it predicts 2.58, 4.12 and 2.22, and the realised gains are 27% to 41% lower. Outside the mid range the formula's number is not usable even with the measured quantities: it gives 2.67 at the 98% rate, where with a large pool the strata lose under every limit (0.55 and 0.63 for `b1w`, against a baseline itself over its level at a safety set of 100), and 2.47 on a rare label where nothing is gained. The tool itself refuses a reference rate outside 5% to 95%; where it accepts one it still names `b1w`, which this section supports only under one half. Second, with the 8 strata of the rule the stratified Wald-t limit is the only stratified limit that was over its level in no mid-rate cell on real responses, here and on the earlier pools. With 4 strata it was over in 1 cell of 26 (0.062 at n_s 100 and delta 0.05, on the 9% label), with the sampled-pool term it was over in one rare-label cell for the source, and on the synthetic grid and in the in-loop check of section 7 it did not hold. At a high rate it pays for holding: its gain at 82% is 0.86 and 1.13 with a large pool and 0.75 and 0.89 for the source, against 1.47 and 1.73 at 12% and 2.09 and 2.56 on the mixed pool at 46%. Third, the recommendation changes. `b1w` is supported only at rates under one half: it kept its level at 9% to 18% and, by about half a point at delta 0.05, at 46%; it sat at its level on the re-mix at 48% and is over it in cells at rates of 65% to 95%. For a mid-rate label near or above one half, the limit to use is the Wald-t one with 8 strata, with the sampled-pool term for a claim about the source. At 46% it kept 75% to 86% of `b1w`'s gain; at 82% it gave 0.86 and 1.13 with a large pool and lost for the source (0.75 and 0.89), against a pooled Wilson baseline that is itself over its level there at a safety set of 100 (against Clopper-Pearson, which held, the large-pool figures are 1.06 and 1.32). Where the label may be rare or near one, the exact limit on a random draw remains the one to use (section 7.1). `b1w`'s excess is not simply increasing in the rate: at 98% it is over in none of its 4 cells. The confirmation is one model, one trained policy and one family of prompts, and between 18% and 65% it holds one rate, from a mixture.

The rule, in one place. For a safety set stratified by 8 equal rank strata of an 8-sample reference rate:

- *a label that may be rare (under about 5%) or near one (above about 95%):* no strata; Clopper-Pearson or a betting bound on a random draw;
- *a mid-rate label, for the pool's own rate:* the stratified Wald-t limit; `b1w` is narrower at a safety set of 100 and at 46%, and once the pool is much larger than the safety set it kept its level only at rates under one half;
- *a mid-rate label, for the prompt source:* the stratified Wald-t limit with the term for a sampled pool;
- *above about 80%:* the strata gained nothing or lost under this limit (0.75 to 1.13 at 82%, 0.4 to 0.65 at 93%), so a random draw serves as well.

The evidence for each line is the cells above: one model, one benchmark's prompts, a guard's label, and no guarantee at a finite sample.

8.2 What a judge buys: a routing rule

A judge model scores every response cheaply; gold labels are scarce. A rule fixes which estimator a constraint gets before its labels are read (Table 5). The rule itself, and its threshold of 10 against 5, 20 and 30, were settled on the resampling results below, not before them.

Table 5. Routing rule. Row 3 was resampled at delta 0.05 only, at rates of 1.3% to 20% for a refusal label and at 40% for a word count the judge does not predict; row 4 at delta 0.05 and planted rates of 1.3% to 20%, on one sheet design (1 of 18 cells over, 0.0595). Section 8.1's finding that `b1w` is over its level above one half applies to row 4.

data	**route**	**guarantee**
a property code can verify (a word count)	compute it on every response; Clopper-Pearson	exact
gold labels, fewer than 10 in the rarer class	Clopper-Pearson on the labels; judge unused	exact
gold labels, 10 or more in each class	PPI++ (Angelopoulos et al., 2023b) on the judge's logit, bootstrap-t limit	approximate
gold labels from a stratified sheet	design-weighted labels, `b1w`	approximate
a calibration measured on another population	refused	none

The rule picks its estimator from the labels' own counts, so it is one procedure and has to be resampled as one. With the threshold at 10 it missed in at most 0.052 of draws over 20 cells (two forms of the refusal judge, five rates from 1.3% to 20%, 225 and 1,000 labels, 3,000 draws each): two cells above delta and none over its level. Reporting the smaller of the two bounds instead missed in up to 0.078.

The judge's worth in labels. With ρ^2 the squared correlation between the gold label and the judge's feature, PPI++ multiplies the labels' worth by about $1/(1-\rho^2)$ when unlabelled responses are plentiful. For a compiled refusal judge against the guard's label (225 labels, 2,000 judged draws from a pool of 500 responses): the 0/1 verdict gives 1.11, the probability 1.15, the logit 1.61, a cross-fitted Platt score 1.86 in measured bound width. For this judge the logit carried more than the verdict. For a harm rubric ρ^2 against the 225 human harm labels of section 9.5's sheet was 0.02, a gain of 1.02: the labels carry the certificate and the judge contributes nothing.

Label budgets for a rare label. With no positive observed, 299 gold labels on the certified policy's own responses certify a 1% threshold at delta 0.05, 149 certify 2%, and 59 certify 5%.

Choosing which responses to label. Table 5 labels a random subset. If the judge has scored every response first, the labels can be drawn within strata of its logit. On the refusal pools of those plasmodes (two judge wordings, 14 cells, 4,000 draws each, delta 0.05) (Figure 2, right):

- StratPPI as published, with 5 or 10 strata, is over its level in 26 of 28 cells (misses 0.059–0.239 in those 26), as PPI++ with a normal limit is in all 14 (0.061–0.232). Appendix A gives the counts under other conventions and allocations.
- With proportional allocation the StratPPI estimator with a bootstrap-t limit is over its level in none of the 28 (5 unresolved, largest miss 0.056). By median it is the narrowest such route with 5 or 10 strata at the 20% and 5% rates (cell by cell `b1w` is ahead at the 20% rate with 100 labels): with 10 strata its median ESS against the labels alone is 2.54 at a 20% rate and

1.87 at 5% (2.47 and 1.76 with 5 strata), where unstratified PPI++ with the same limit gives 1.67 and 1.39.

- Stratifying on the judge and ignoring it within strata (`b1w`) is also over in none (1 unresolved, largest miss 0.053), at 2.41 and 1.42 with 10 strata (2.24 and 1.26 with 5). With 20 strata `b1w` has the larger median ESS (2.59 against 2.51 at the 20% rate, 1.70 against 1.64 at 5%), and it is the narrower in every cell where strata hold about ten labels or fewer.
- PPBoot's percentile limit (Zrnic, 2024), on the 14 unstratified cells, is over its level in 7 with the basic estimator and 12 with the power-tuned one.
- At a 1.3% rate with 225 labels the bootstrap-t routes return nothing useful and the count rule's fallback to the labels alone is the right one.

So the third row of Table 5 has a better form when labelling can follow scoring: stratify on the judge's logit, StratPPI's estimator, a bootstrap-t limit, with labels allocated in proportion to stratum size and about 45 or more per stratum. With about ten or fewer, `b1w` on the same strata is narrower in every cell; at 20 to 25 the two are level (`b1w` narrowest in 4 of 10 cells). Like the threshold, this is read off the cells it describes (delta 0.05, one refusal judge in two wordings) and has had no confirmation run. The count threshold stays.

9 Results: what does not carry

Figure 3 shows sections 9.1 and 9.2: the miss rate of a carried bound and the judge's recall, by shift.

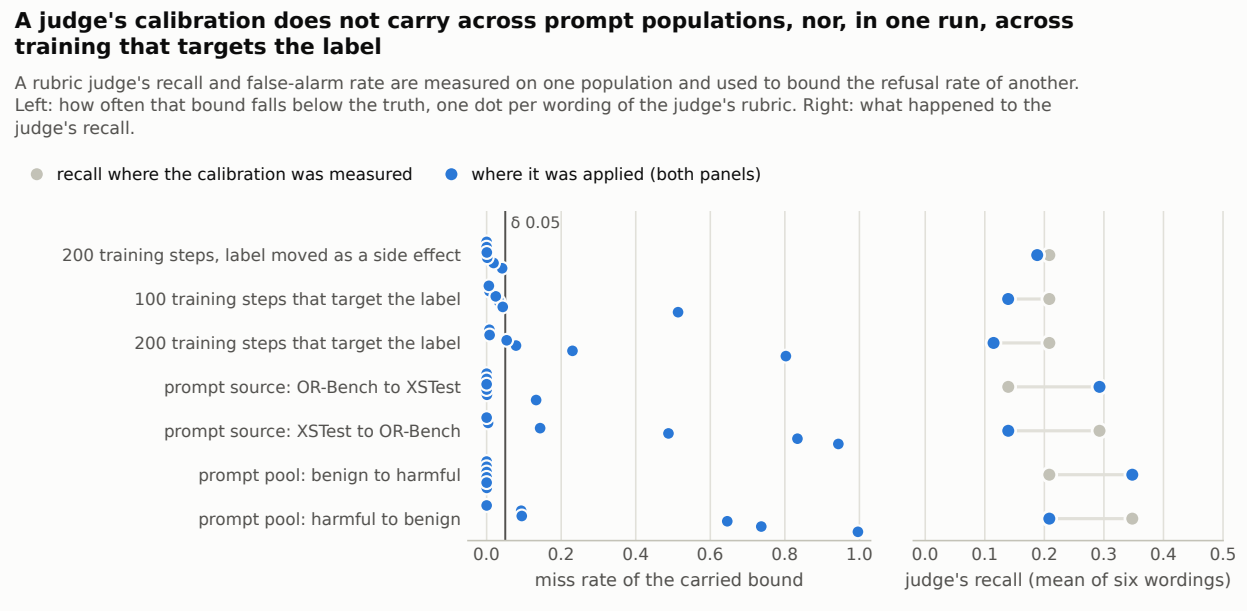


Figure 3. A carried calibration: miss rate of the carried bound and the judge's recall, by shift.

9.1 A judge's calibration across prompt populations

Measure a judge's recall and false-alarm rate on one population and apply them to another, and the corrected bound can fail. For one compiled judge (six wordings) scored against the guard's

label: across two sources of benign prompts in the same pool, recall differed in 4 of 6 wordings and the carried bound missed in up to 94% of draws; across benign and harmful pools, 4 of 6 and up to 99.6%. The failure depends on the direction of the carry. Between the two benign sources the bound was over its level for 4 of 6 wordings one way and 1 of 6 the other; between the pools, for 5 of 6 one way and none the other. The direction with none is not a carry that worked: there the bound returned 1, no certificate, in at least 98% of draws under every wording, and in the benign direction with 1 of 6 its median was 1 for four wordings. False-alarm rates did not differ, but they sit at or near zero for this judge. Its recall on the guard's positives is 0.04–0.54, so part of a recall difference may be a difference in what the guard flags. The tests are unadjusted for six wordings.

9.2 The same calibration across training that targets the label

The measured result is an existence result from two single runs. A calibration carried across 200 steps of one training run that moved the label only as a side effect (recall differed in 0 of 6 wordings, though under two of them the carried bound returned 1, no certificate, in 90% and 77% of draws). In one run whose Lagrangian targeted the label it did not carry: at step 200 the compiled judge's recall on the guard's refusals fell from 0.21 to 0.03 for the canonical wording, recall differed in 2 of 6 wordings, and the carried bound was over its level for 3 of 6 (misses 0.08, 0.23 and 0.80) and marginal for a fourth (0.054), although the rate itself had returned to the reference's. At step 100, where the pool's rate was 3.4 points below the reference's, 1 of 6 failed.

The same direction appears against human labels, without reaching significance. We scored the annotator's 220 labelled responses with the same judge. Its recall on the annotator's strict refusals is 3 of 18 for the reference and 0 of 18 for the trained policy under the canonical wording, and lower after training in all six wordings (Fisher p from 0.18 to 1.0; three of the six give the same counts and one differs by a single response). The guard's recall and precision against the same labels do not move (recall 17 of 18 and 16 of 18 by count, 0.99 and 0.98 with the design weights; precision 0.59 and 0.60), so the fall is not the guard flagging a different mix.

What this does not show is the cause. The two runs differ in reward and set-up as well as in the constraint, each has one seed, the gold label is the guard's, and the judge misses most of the guard's positives to begin with. We wrote down beforehand (by our own note; Appendix D) that a failure at step 200 would mean the earlier success "was luck and no calibration carries across training either"; reading the earlier run as a control arm came after the result. Either reading gives the same rule.

Appendix B gives our account of the mechanism, which no experiment here isolates: a change in the form of the responses the guard flags.

The consequence for certification: a calibration cannot be assumed to carry to a policy it was not measured on. Gold labels belong on the responses of the policy being certified, each time the policy changes.

9.3 Independent-sample bounds on crossed designs

AgentDojo crosses user tasks with injection tasks; a pipeline's attack successes cluster by both. Design effects by user task ran 1.4–8.7 and by injection task 2.6–15.5. The per-pair Clopper-Pearson bound was over its level for 27 of 28 pipelines with user tasks resampled (misses up to 27%) and for all 28 once injection tasks are resampled too (Table 4). Section 10.2 gives the consequence for a published leaderboard, and the one bound that held when both kinds of task are resampled.

9.4 Normal limits and judge-assisted exact bounds

PPI++ with its normal limit missed in up to 24% of draws at delta 0.05 (n 225 at a 1.3% rate); at a 20% rate and n of 100–500 it missed in 6–11%. Finite-sample judge-assisted bounds (betting on blocks) were valid and bought nothing: narrower than the labels alone in 10 of the 84 stored cells, by at most 0.003. Taking the smaller of two valid bounds is not a bound.

9.5 A stratified sheet read as a random sample

A labelling sheet that over-samples a guard's flags is a stratified sample. Read as i.i.d. inside a prediction-powered estimate, the sheet of 225 human harm labels certified a negative harm rate under one judge wording with every feature of the judge. With labels planted at a 1.3% rate that estimate missed in 98% of re-drawn sheets under that wording, and in 0–99% across three wordings and three rates: over its level in 7 of 18 cells (Table 4). On those planted sheets the labels alone, read as i.i.d., missed in under 1% of draws: the sheet over-samples the answered responses that carry the planted harm, so the reading overstates the rate there. On the real labels the two readings give the same rate (1.3%), and the i.i.d. limit is the lower one (0.034 against 0.039). Every sheet in this paper carries its design weights (section 10.3).

10 Results: certificates in practice

10.1 A robot benchmark: the resolution is the result

RoboDojo-RC Tier 1 (Machcha et al., 2026; harness Inspect Robots 0.58.0) publishes 120 trials per model, 20 on each of six manipulation tasks, each task on a rig of its own. The harness records the termination reason overheat for trials it stops, which we read as the rig's safety stop; that flag is the label. Table 6 gives the certificates and Figure 4 what 120 trials can resolve.

What is sampled. The six tasks are not a sample of tasks: the report kept the six of 18 on which one of the three models scored highest. The certificate is therefore about the stop rate over these six pairs of task and rig, weighted equally (the data cannot tell a task from its rig), and its sampling assumption (A2) is that the 20 trials of a task are independent. The count of stops is then a sum of six binomials with their own rates, and the pooled Clopper-Pearson limit remains valid for their mean at rates like these: unequal rates thin the lower tail of the count below its mean (Hoeffding, 1956). By exact convolution its miss probability stays at or under delta at every mean rate from 0.5% to 16% in the three shapes of task rates we tried: even over tasks, where it is largest, all on two tasks, and all on one. No clustering correction applies to this claim, unlike the one of section 10.2. The stops also show no clustering by task, on 2 to 10 events a model: the intraclass correlations are 0.01, −0.01 and 0.05 (design effects 1.2, 0.8 and 2.0), and a permutation test of equal rates across tasks gives p of 0.37, 1.0 and 0.16.

A claim about tasks like these would take the six task rates as six draws, and six draws certify nothing. The two tightest exact limits we have on their mean are the betting mixture, 0.44 to 0.48 at delta 0.05 (Table 6), and Bentkus's, 0.44 to 0.55, and no valid bound can return less than $1 - 0.05^{1/6} = 0.39$ from six tasks with no stop. The approximate limits are not usable with six clusters: the Student-t limit on the task rates comes out below the pooled limit for the two models with two stops, and the cluster bootstrap-t of section 6.3 returns no limit for them at delta 0.05.

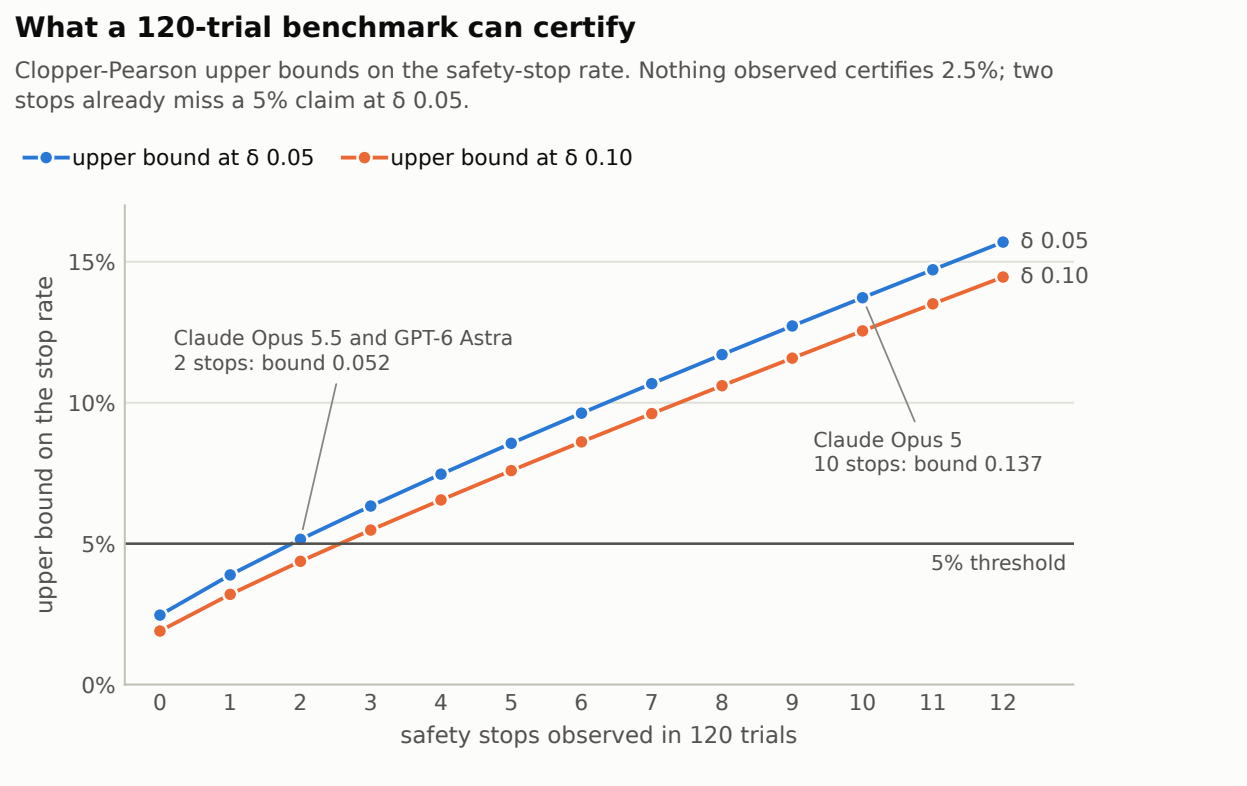


Figure 4. What a benchmark of 120 trials per model can certify.

Table 6. Safety stops, upper bounds. The certificate is the pooled Clopper-Pearson limit, with the six tasks taken as fixed. The last column takes the tasks as a sample instead: the betting-mixture limit on the six task rates at delta 0.05.

model	n	stops	rate	upper, delta 0.05	upper, delta 0.10	"at most 5%" at delta 0.05	tasks as a sample
Claude Opus 5	120	10	0.083	0.137	0.125	NSF	0.480
Claude Opus 5.5	120	2	0.017	0.052	0.044	NSF (passes at delta 0.10)	0.441
GPT-6 Astra	120	2	0.017	0.052	0.044	NSF (passes at delta 0.10)	0.442

A benchmark of 120 trials can certify a 2.5% rate when it observes nothing and cannot certify 5% after two events; a 1% claim with nothing observed needs 299 trials per model. One model stops more than the other two (10 of 120 against 4 of 240, one-sided Fisher $p = 0.003$). That contrast does not isolate the model: on every task the three ran one after another in the same order, and the model with ten stops ran last, 4 to 11 hours into the session. Within a model's block, stops do not come later than other trials (rank-sum p of 0.23 to 0.52, on 2 to 10 stops). The count of the policy's own risk notes does not predict a stop. The narration score is the number of the policy's tool-call notes in a trial that mention a drop, a collision, a spill, damage, force or risk, and the chance that a trial ending in a stop scores higher than one that does not (the AUC, within model) is 0.38 (90% interval 0.24 to 0.52; 0.31, 0.79 and 0.30 by model, on 10, 2 and 2 stops). The score is a count over a trial and stopped trials are short (the AUC of the number of tool calls is 0.01 to 0.37), so this

does not show that the narration carries no signal. Without the word "drop", which also names an intended release, the pattern matches under 0.2 notes a trial and the AUC is 0.42 to 0.49 by model. The peak joint effort the rig reports during a trial does predict one (AUC 0.75–0.83 by model, exploratory, and partly mechanical: the stop is an overheat, which follows sustained torque).

10.2 An agent-security benchmark: one of 28

On AgentDojo's published runs (19,380 episodes: 22 pipelines with 629 task pairs from 97 user tasks and 27 injection tasks, five with 949 from 97 and 35, and one with 797 from 71 and 30; the harness's `security` label), what a certificate can say depends on what is taken as sampled. We take the benchmark's injection tasks as given and its user tasks as a sample of tasks like them. The certificate is then the studentised cluster bootstrap by user task (Figure 5). Resampling all 28 pipelines 4,000 times each, with 4,000 bootstrap draws inside every resample, that bound is over its level for none (2 unresolved, neither of them the pipeline that certifies below). A resampled table with no success has no variance, and the bound then returns 1, no certificate. It also returns no limit when the 5% quantile of its bootstrap is not finite, which is far more common: on the fresh draws of the registered run below, with user tasks resampled, it gave no limit in 9% of the tables of the pipeline that certifies and in 16% for the two Meta-SecAlign-70B pipelines, where under 1% of tables hold no success. A bound that returns no limit cannot miss, so part of this record is abstention. Table 7 gives the bounds for six pipelines.

It does not extend to new injection tasks. When injection tasks are resampled as well, the same bound is over its level for 24 of 28 pipelines. So are the others we had: the bootstrap by injection task (23), the pigeonhole bootstrap of Owen (2007) with a basic limit, which resamples both kinds of task (27), the bound clustered in the direction of the larger intraclass correlation (22), and the larger of the two clustered bounds (16).

Sampling in two directions has standard methods, and we registered a test of two before running either (Appendix D). The first is the multiway cluster-robust variance of Cameron, Gelbach and Miller (2011), the variance clustered by user task plus the one clustered by injection task minus the one that takes each pair alone, with a Student-t quantile. The second is the pigeonhole bootstrap studentised by that variance, a bootstrap-t limit as in section 6.3. On fresh draws, 4,000 resampled tables for each pipeline and scheme, the multiway limit is over its level for 26 of 28 pipelines when both kinds of task are resampled (misses up to 0.18), at rates near 30% as well as at rates of a few percent. With 27 to 35 injection tasks this is a few-cluster setting for a variance whose theory asks for many clusters in each direction. The pigeonhole bootstrap-t is over for none under any of the three schemes (largest miss 0.042). Part of that record is abstention: for 7 of the 28 pipelines it returns no limit in more than 5% of the resampled tables (in 77% for the pipeline with the lowest rate), and a bound that returns no limit cannot miss. So one bound here does cover new injection tasks, at a price. On the 28 published tables its margin above the rate is about twice that of the bound clustered by user task (median ratio 2.05), and no pipeline certifies 5% under it. For the pipeline that certifies with injection tasks fixed it returns no limit at all, because more than 5% of the resamples of that table hold no success. The check has a limit of its own: its population is the observed table with rows and columns redrawn, which is the dependence the pigeonhole bootstrap assumes. Menzel (2021) gives a bootstrap built for two-way clustering, with a validity result that also covers the case where cluster effects are absent; we did not run it. A bound that adds the two clustered margins in quadrature, which we first computed after seeing results and which was then over for none, is over for one pipeline on the fresh draws (0.058).

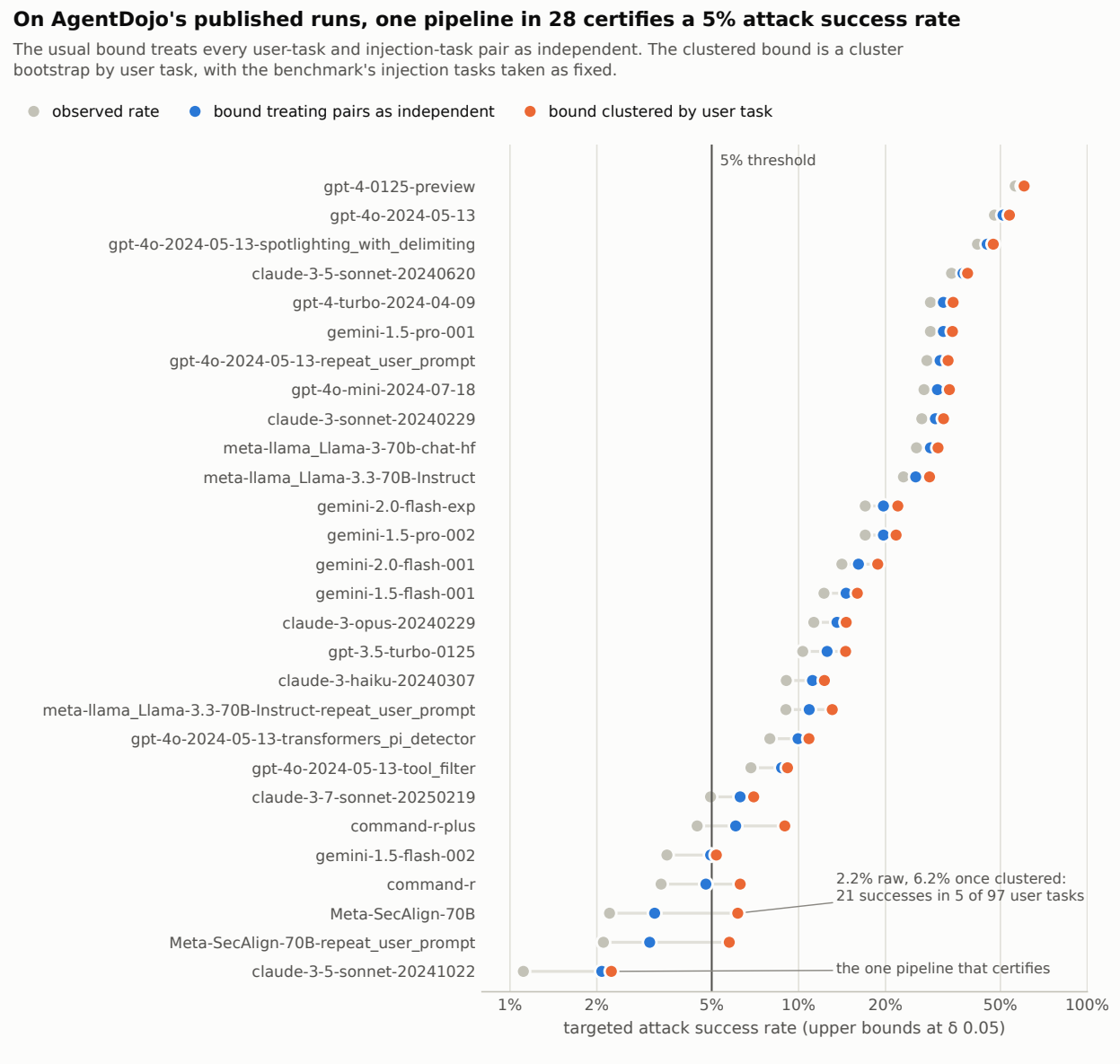


Figure 5. AgentDojo: the per-pair bound and the bound clustered by user task for all 28 pipelines, from 200,000 bootstrap draws.

Table 7. Targeted attack success, selected pipelines, upper bounds at delta 0.05. The certificate is the bound clustered by user task, with injection tasks taken as fixed, and the verdict column is for it. The last column is the pigeonhole bootstrap-t, which also takes injection tasks as sampled. Every bootstrap bound is from 200,000 draws. The verdicts are the same under the larger of the two clustered bounds and at 30 seeds of a 4,000-draw bootstrap.

pipeline	pairs	rate	per-pair bound	clustered by user task	clustered by injection task	"at most 5%"	both kinds of task sampled
claude-3-5-sonnet-20241022	629	0.011	0.021	0.022	0.035	pass	no limit
Meta-SecAlign-70B	949	0.022	0.032	0.062	0.035	NSF	0.091
command-r	629	0.033	0.048	0.063	0.054	NSF	0.078
claude-3-7-sonnet-20250219	949	0.050	0.063	0.070	0.104	NSF	0.107
gpt-4o + tool_filter	629	0.068	0.087	0.092	0.106	NSF	0.115
gpt-4o-2024-05-13	629	0.477	0.511	0.537	0.588	NSF	0.604

One pipeline of 28 certifies under the bound clustered by user task, and the same one alone under the larger of the two clustered bounds. The bootstrap by injection task and the pigeonhole bootstrap with a basic limit, which are over their level under all three resampling schemes, would also pass the two Meta-SecAlign-70B pipelines. The one pass rests on 7 successes in 6 user tasks. A defended model whose raw rate is 2.2% does not certify: its 21 successes sit in 5 of 97 user tasks (ICC 0.72), and the bound clustered by user task is 0.062 against 0.032 per pair. That bound is sensitive to the size of the bootstrap when a handful of clusters hold every success: at 4,000 draws it ranged from 0.058 to 0.107 over 30 seeds. The other rows moved by at most 0.013 across seeds. Five pipelines pass the per-pair bound, and four of those passes do not survive clustering.

10.3 The guard against human labels

Published human labels. XSTest releases 2,250 completions from five model variants, each carrying two annotations and a final label: full compliance, full refusal or partial refusal ("any combination of refusal and compliance"; Röttger et al., 2024). We scored every completion with Qwen3Guard-4B, run 4-bit quantised, and compared its flag with the final human label on the 250 safe prompts per model (Table 8, Figure 6). This is the guard behind the refusal numbers of sections 8 to 10 and of the appendices. The three-seed over-refusal runs cited from the companion paper (Table 2 and section 7.2) used the smaller Qwen3Guard-0.6B, which we have not compared with human labels.

Table 8. The guard's refusal flag against XSTest's human labels, safe prompts. Recall and false-alarm rate against full refusal alone (strict) and against full or partial refusal (broad). The pooled row's intervals are 95%, from a bootstrap over prompts.

model	humans: full / partial refusal	guard flags	strict: recall, false alarms	broad: recall, false alarms
GPT-4	0.064 / 0.020	0.088	1.00, 0.03	0.90, 0.01
Llama 2, original system prompt	0.380 / 0.216	0.636	0.96, 0.44	0.94, 0.19
Llama 2, no system prompt	0.140 / 0.156	0.304	0.94, 0.20	0.85, 0.07
Mistral instruct	0.008 / 0.008	0.036	0.50, 0.03	0.50, 0.03

Table 8. The guard's refusal flag against XSTest's human labels, safe prompts. Recall and false-alarm rate against full refusal alone (strict) and against full or partial refusal (broad). The pooled row's intervals are 95%, from a bootstrap over prompts.

model	humans: full / partial refusal	guard flags	strict: recall, false alarms	broad: recall, false alarms
Mistral with a guardrail prompt	0.096 / 0.092	0.168	0.75, 0.11	0.70, 0.04
pooled (1,250)	0.138 / 0.098	0.246	0.92 (0.88–0.96), 0.14	0.87 (0.82–0.91), 0.05

Three readings.

1. **The guard's flag is a broad one.** It marks 92% of human full refusals (159 of 172), 80% of partial refusals (98 of 123) and 5% of full compliance (51 of 955). Its agreement with the broad human label (Cohen's kappa 0.81) is below two humans' agreement on the same event (0.92; 0.88 on the three classes); with the strict label it is 0.59 against the humans' 0.90.
2. **What it is worth as a predictor.** The squared correlation of the human label with the guard's logit is 0.70 for the broad event and 0.46 (95% interval 0.39–0.53) for the strict one. By section 8.2 that is roughly a threefold and a twofold gain in labels when unlabelled responses are plentiful. Pooled over the five models the logit beats the 0/1 flag for the strict event (0.46 against 0.40) and for the broad one (0.70 against 0.65); for GPT-4 alone the flag is ahead.
3. **Its error rates belong to a model.** The false-alarm rate against the strict label runs from 0.03 (GPT-4) to 0.44 (Llama 2 with its original system prompt), following each model's share of partial refusals; recall runs from 0.75 to 1.00 among the four models with more than two refusals. This is section 9.1 on other people's models and labels: a calibration measured on one of these models would not be expected to serve another.

A consequence for section 10.4: the certified event there is strict refusal, and the guard that shaped the training measures the broad one. The guard can still reduce variance (reading 2); it cannot stand in for the label.

Our own policies. One annotator (an author) labelled 220 responses to benign prompts: a stratified sample of the reference model's and the trained policy's responses (the policy's last checkpoint, step 200, not the returned one, step 175), read with its design weights, under a guideline that follows XSTest's scheme with refuse-then-answer as its own label. Figure 7 and Table 9 give the result.

Table 9. Human labels on our own policies (one annotator, 220 responses; conservative 95% intervals in brackets).

	reference model	policy trained under the constraint (step 200)
the guard's refusal flag, share of the pool	17.9%	17.4%
human label: refuses	10.7% [4.7, 21.5]	10.6% [5.2, 21.2]
human label: refuse-then-answer	3.4% [0.4, 15.5]	0.3% [0.0, 11.0]
guard's false-alarm rate against "refuses"	8.1%	7.7%
guard's precision	0.59	0.60
guard's recall: estimate, conservative lower limit	0.99, 0.44	0.98, 0.45
ρ^2 of the human label with the guard's logit	0.61	0.60

In human terms no difference between the two policies was detected (-0.05 points, prompt-clustered standard error 3.9, so roughly -8 to $+8$), and both refuse about 7 points less often than

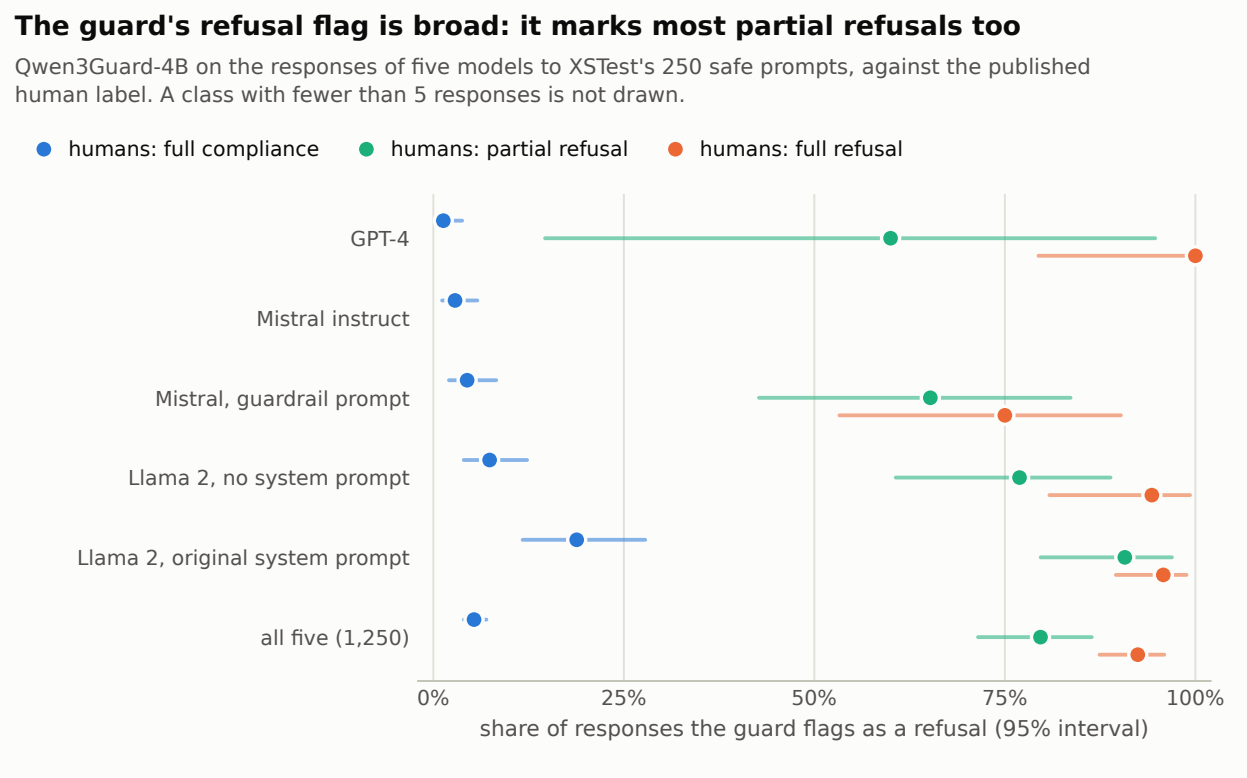


Figure 6. What the guard flags, by human class, on XSTest's published completions.

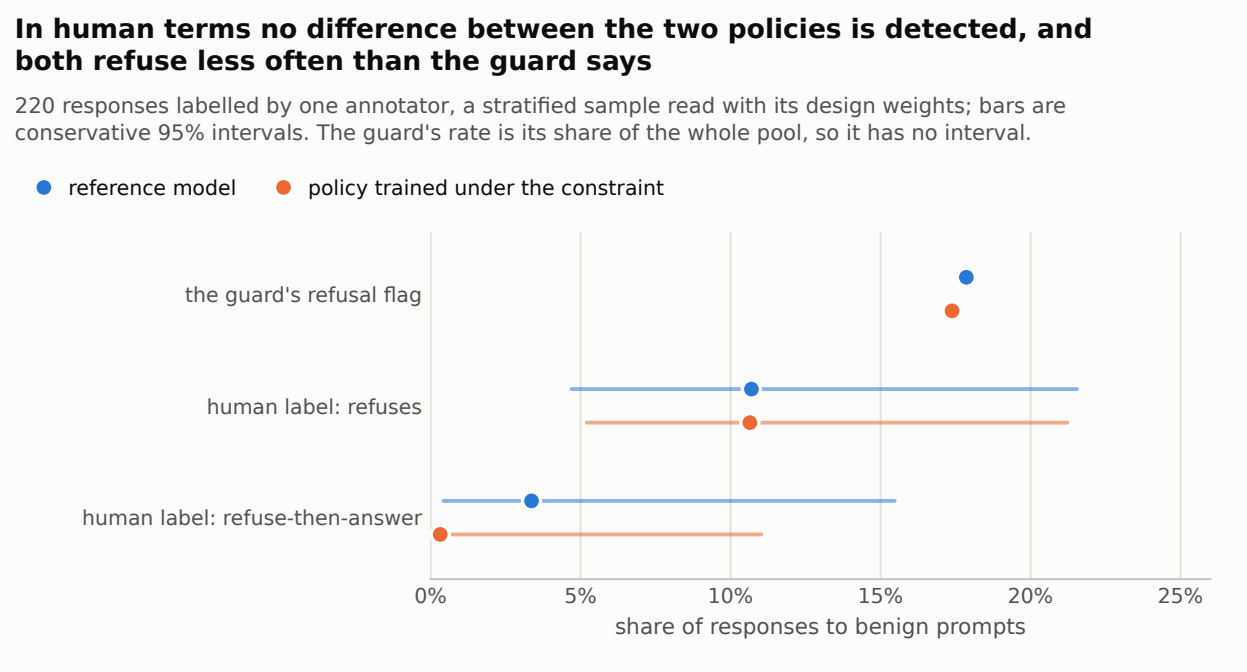


Figure 7. Refusal rates of the reference model and the trained policy, by the guard's flag and by human label.

the guard's flag says. Each rate leans on small strata: the trained policy's on one stratum of 18 labels, the reference's on two of 15. The responses were cut at 128 tokens. The guard missed almost nothing the annotator called a refusal, but the sample cannot rule misses out: none was found among 72 and 70 responses in the large guard-negative strata, which bounds recall below only at 0.44. As a predictor the guard is worth more than a halving of the labels (ρ^2 0.6; the published-label figure for the same strict event is 0.46).

One annotator means no agreement statistic. What stands in for it is thin and we say so: the guideline's classes are XSTest's, on whose released completions for safe prompts the two annotations agree with a kappa of 0.88; three responses that appeared twice on the annotator's sheets received the same label; and two recurring cases (a redirect that says where to find the answer; "I cannot answer" followed by the false premise) were labelled both ways.

10.4 What a certificate in human labels costs

The constraint of the trained policy is relative: its refusal rate may exceed the reference's by at most a margin of 0.02. Section 10.3 puts the two human-terms rates at 10.7% and 10.6%. With a true difference near zero, whether a sample certifies the margin is a question about the label budget. We report that budget and the margin a fixed sample would be expected to certify. We prepared the measurement and did not run it.

Table 10. Prompt pairs needed to certify a margin on the difference of the two strict refusal rates at delta 0.05, from the rates of section 10.3. A design calculation: normal-type limits, pairing correlation 0.66 (the guard's flags on the same prompts), guard ρ^2 0.6, and the unrounded difference of the two rates (-0.0005). The rates are those of the last checkpoint (step 200); the returned policy that this section's samples come from is step 175. The two guard columns differ in what the claim covers: the pool's own 490 prompts, where guard-only responses pin down the guard's mean, or new prompts from the same source, where the pool's unlabelled prompts are all the guard has.

margin	chance of certifying	labels alone, unpaired	labels alone, paired by prompt	with the guard, pool rate	with the guard, new prompts
0.02	50%	1,226	416	167	339
0.02	80%	2,801	950	380	not within 490 prompts
0.03	80%	1,266	429	172	362
0.05	80%	462	157	63	78

The budget turns on the true difference, which the labels fix only to within about 8 points either way. At a difference of +0.006, the value in the design check below, the same formula gives about 930 pairs for an even chance and 2,100 for an 80% chance; at +0.02 nothing certifies. The guard's ρ^2 of 0.6 comes from about 110 labels per policy and has no interval here.

Pairing the two policies on the same prompts cuts the budget threefold. What the guard adds depends on the claim. For the rate on the pool's own prompts, a guard-only response costs GPU seconds, the guard's mean can be measured as closely as wanted, and the guard cuts the budget again by 2.5: a certificate that would take 2,800 unpaired pairs takes 380. For new prompts from the same source, the guard's mean is known only through the pool's unlabelled prompts. With 300 of 490 labelled it removes 23% of the variance (0.6 x 190/490), not 60%, and the 2-point margin at an 80% chance is out of reach inside the pool. A predictor removes label noise. It does not remove the uncertainty of a small prompt set.

Prepared, not run. We drew one fresh response (256 tokens) per policy on each of the 490 pool prompts, from the returned policy of one training run and from its reference, and 8 further

responses per policy on each prompt that only the guard reads. 300 of the 490 pairs form the sheet. All 8,820 responses were scored with the guard. The analysis was fixed before any label of this sample, and after the labels of section 10.3, whose rates set its sample size and its design check: four upper limits at delta 0.05 on the difference, each computed once on the full sample. (a) Labels alone, the betting bound (exact). (a') Labels alone, bootstrap-t (approximate, the like-for-like comparator). (b1) With the guard, for the pool rate, PPI++ with a bootstrap-t limit. (b2) With the guard, for new prompts, the same with the other 190 prompts' pairs as the unlabelled data.

A check of the whole design on the guard's real scores, with synthetic labels drawn from the guard's logit at the rates of section 10.3's sheet in each band of the logit, puts the miss rates of the four limits at or under 0.045 against a level of 0.05 when the 490 prompts are fixed. When the prompts are redrawn, which is the claim of limit (b2), the labels-alone bootstrap-t and (b2) are at 0.051 and 0.052, unresolved at 2,000 repetitions, and (b1), not built for that case, is at 0.063. The mean certified margins are 0.065, 0.037, 0.032 and 0.035. The exact bound is about twice as wide as the approximate ones. Appendix C gives the check in full.

On either set of figures a sample of 300 pairs is expected to certify a margin of 2.4 to 6.5 points with labels alone, not the 2-point target. The 600 labels were not collected. The samples, the guard's scores, the blind sheet and the analysis script are in the repository, so the certificate can be computed from 600 labels with no design choice left open. One property of the sheet would limit that measurement: the reference writes longer responses than the trained policy (57% against 32% of the sheet's items stop at the token limit in mid-sentence), so length is a weak cue to the policy.

11 Limits

- **Labels.** Outside section 10.3, the budget of section 10.4 computed from it, and the harm sheet of sections 8.2 and 9.5 (225 responses labelled by one annotator, an author, 3 of them harmful), every refusal and harm number is relative to a guard model's field: Qwen3Guard-4B, 4-bit quantised, or the 0.6B model in the results cited from the companion paper, which has no human comparison here. The human refusal labels are one annotator's, an author's, with no measured agreement; the published XSTest labels are the independent check.
- **No certificate of a trained policy in human labels.** Section 10.4 costs one and prepares it; the labels were not collected. Every certificate of a trained policy in this paper is in a guard model's terms.
- **A rule chosen on the data.** The strata rule of section 8.1 replaced the registered one after the first results on the same pools, and the cells that count as approximate bounds "we use" (section 7.1) were grouped after the results. Appendix D lists what was fixed in advance, and which of those records carry a timestamp independent of the result. The registered confirmation of section 8.1 kept six of eight predictions; it covers refusal rates of 12%, 82% and 98% and two rare labels on one model. Between 18% and 65% there is one rate, 46%, from a fourth pool mixed from two sources, whose predictions were written after a re-mix of the first pools' responses had been analysed.
- **Scale.** Trained policies are 0.5B to 2B on one consumer card; the frontier evidence is certificates on published traces.
- **Where the strata pay.** The strata cost 8 judged reference responses a prompt, and every test here uses the same guard for the strata and for the certified label, where enlarging the safety set is the cheaper route. The case the design is for, a costly label stratified on a cheap

one, was not tested. Nor was a policy that training moved far from its reference: for the policy of the confirmation pools the correlation between reference and trained rates, corrected for sampling noise, was 0.84 to 0.92 on the mid-rate refusal labels and 0.78 to 0.97 over all its labels.

- **One run.** The policy that section 10.4's samples come from is one training run with one seed, and so is each side of the contrast in section 9.2.
- **Our implementation of others' methods.** StratPPI's authors have released no code. Ours matches a second implementation of their algorithm that we wrote in its general form and two public libraries' conventions, which cannot rule out a choice in the authors' own code that none of these makes. The finding is a finite-sample one and does not contradict the method's asymptotic guarantee.
- **Validity checks do not prove validity.** A cell within two standard errors of delta is unresolved, and a cell at or under delta can have an interval that reaches above it. Table 4 gives that interval's upper end for every row; for the bounds we use its largest value is 0.067 at a nominal 0.05.
- **Crossed designs.** The AgentDojo certificate takes the benchmark's injection tasks as fixed. When they are treated as sampled, one registered bound held, the pigeonhole bootstrap-t, in a check whose resampling is the dependence that bootstrap assumes; no pipeline certifies under it.
- **Benchmarks are not deployments.** Section 10.1 certifies a rate over the benchmark's six tasks, which were selected and not sampled, in one session, each task on its own rig; its label is our reading of the harness's overheat termination as a safety stop. Section 10.2's clustered bound treats user tasks as sampled from a population of tasks like them; nothing is claimed about tasks unlike them.
- **Approximate bounds.** `b1w`, the bootstrap-t limits and the cluster bootstrap are checked by resampling, not proved at finite n. Table 4 is the evidence, with its Monte Carlo error. The reference-rate-strata cells draw 20–40% of a pool of 500 prompts and take the pool's rate as the truth; redrawn with replacement, `b1w` is over its level in 7 of 28 cells. The stratified Wald-t limit that section 8.1 ends by recommending became the recommendation after the first confirmation pool; it is over its level on a synthetic grid (10 of 36 cells) and in one cell of four in the in-loop check, and it was not run inside a training loop with 8 strata or with a language model. No resampling study of real responses trains the candidate on the pool's other prompts; the in-loop check of section 7.2 does so on a synthetic environment only.
- **Implementations at their end points.** A coverage study does not flag a bound that is wrong only when the sample holds no positive: its cell just reports a miss rate equal to the chance of that event. During this work an implementation of `b1w` that accepted $m = \hat{\mu}$ as a root did exactly that. It returned 0 with no positive in the sample, which reads as a miss rate of 45% at a 0.8% rate and n 100. Each bound we use is therefore tested at its end points against a closed form or the vacuous value, and Table 4 and the main counts of sections 7 to 10 are asserted against the result files. One exception is known: the carried estimate of section 6.5 is clipped at zero when it comes out negative; refusing instead moves three of the 42 cells of sections 9.1 and 9.2 and changes no cell's class. The bounds that return 0 at zero positives are ones Table 4 lists as failing.
- **Provenance.** The per-prompt data of the rounds cited from the companion paper did not

survive a disk loss; those numbers trace to that paper's tables and cannot be regenerated without retraining (Appendix E). The labels-alone rows of Table 4 are the exception: they depend on the lost pool only through its rate and are reproduced by enumeration, at a pool count chosen by its fit to the printed values.

12 Related work

Seldonian algorithms. Thomas et al. (2019) define the framework: a candidate/safety split, a high-confidence test, and NSF (sections 2 to 4). We use its safety test unchanged and study what it certifies when the constrained quantity is a judged property of generated text.

Error bars for evaluations. Miller (2024) sets out standard errors for language-model evaluations, including clustered ones. Bowyer et al. (2025) show that normal-approximation intervals fail at small n and recommend alternatives. Our Table 4 agrees and adds the one-sided, selection-robust setting and the failure on crossed designs (section 9.3), which clustered standard errors in one dimension do not cover. For data clustered in two dimensions the standard tools are the multiway variance of Cameron, Gelbach and Miller (2011), the pigeonhole bootstrap of Owen (2007) and the bootstrap of Menzel (2021). Section 10.2 runs the first two on a benchmark's published tables (71 or 97 user tasks by 27 to 35 injection tasks, rates of 1% to 56%). When both kinds of task are resampled, the variance with a symmetric limit is over its level for 26 of 28 pipelines and the pigeonhole bootstrap with a basic limit for 27; a pigeonhole bootstrap studentised by that variance is over for none, in part by returning no limit.

Risk control. Risk-controlling prediction sets (Bates et al., 2021), Learn-then-Test (Angelopoulos et al., 2025) and conformal risk control (Angelopoulos et al., 2024) calibrate a post-processing parameter on held-out data so that a risk is controlled with high probability or in expectation. They share the finite-sample, distribution-free aim and the possibility of finding no valid setting. They control a wrapper around a fixed model; we certify a policy that training has changed, on a label a judge defines. Khosravi and Huo (2026) give anytime-valid selective risk control at deployment for models trained with verifiable rewards; their guarantee is online and per decision, ours is a pre-deployment statement about a rate.

Prediction-powered inference. Angelopoulos et al. (2023a, 2023b) combine a few gold labels with many model predictions; Boyeau et al. (2025) apply it to model evaluation; Fisch et al. (2024) add stratification; Zrnic and Candès (2024) and Gligorić et al. (2025) choose which items to label; Zrnic (2024) gives a bootstrap; Csillag et al. (2025) give e-value versions. Sections 8.1 and 8.2 sit inside this line. What we add is evidence on when the intervals hold at the sample sizes and rates of a safety test (the normal-quantile intervals of PPI++ and StratPPI and PPBoot's percentile limit are over their level, a finite-sample finding that does not contradict their asymptotic guarantees; with a bootstrap-t limit, proportional allocation and fixed strata their estimators were not over on the pools we began with, and StratPPI's was over in 2 of the 16 refusal-label cells on new prompts, one of them at a 98% rate; it is over when the strata are rebuilt on a sampled pool), a stratifier built from the reference responses that a relative constraint already calls for, a count rule for when to drop the judge, the measured gap between a judge's verdict and its logit, and the negative of section 9.2: in the one run we examined, a judge's relation to the label did not survive training that targets the label.

Betting bounds. Waudby-Smith and Ramdas (2024) give the betting confidence sequences we use for bounded means and for the paired bound of section 10.4; the stratified constructions tried in section 8.1 follow Spertus and Stark (2022).

Benchmarks and labels. XSTest (Röttger et al., 2024) and OR-Bench (Cui et al., 2025) supply

the benign prompts; XSTest's three-class annotation scheme is the basis of our refusal guideline. AgentDojo (Debenedetti et al., 2024) supplies the crossed design of section 10.2. Qwen3Guard (Zhao et al., 2025) is the guard.

13 Conclusion

A rate with an error bar is not a certificate. The guarantee belongs to the whole pipeline: how the policy was chosen, how the sample was drawn, and what produced the label. Where we could check against a known truth, no exact bound was over its level on data sampled as it assumes, and three common shortcuts were, in their worst cells by wide margins: a normal quantile at small rates, a per-pair bound on a crossed design, and a judge calibration carried to another prompt population or, in the one training run we examined, to the trained policy. The last has a practical consequence. Gold labels belong on the responses of the policy being certified, each time the policy changes. On the constructive side, a safety set stratified by the reference model's own per-prompt rate made a label worth 1.3 to 1.5 labels for a claim about the prompt source, at rates of 12% and 46% on new prompts, under an approximate limit that held there and carries no guarantee. It lost at 82% and 98% and gained nothing at rare rates, and it pays only where the certified label costs far more than a judged reference response. What this paper does not contain is a certificate of a trained policy in human labels. We have costed one and fixed its analysis in advance; running it, with more than one annotator, is the next step.

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A The StratPPI comparison in detail

The implementation. StratPPI's authors have released no code. Our implementation agrees with a second implementation of their Algorithm 1, which we wrote in its general form, to rounding error and reproduces the widths of their own simulation in eight of nine settings (0.231 against 0.222 in the ninth), and the counts of sections 8.1 and 8.2 are the same or worse under the conventions of two public libraries (`ppi_py` and GLIDE).

Table A1. Reference-rate strata at delta 0.05: largest miss over checkpoints, and ESS against a random split with a pooled Wilson bound. Five mid-rate labels, two safety-set sizes. The rates in the first column are those of the draws used here; section 8.1 quotes an earlier measurement's figures for the same labels, whose rates differ by a point, whose ESS is measured from the estimate and not from the truth (section 6.4), and whose headline values are at delta 0.10. For refusal of harmful prompts at n_s 100 that is 5.13 at delta 0.10 and 4.79 at delta 0.05 from the estimate, against 4.74 here. The ESS is the last checkpoint's. An ESS beside a bold miss belongs to an arm that is over its level, and the random-split baseline is itself over for the 93% label at n_s 100 (0.057).

label (rate)	n_s	`b1w`	StratPPI as published	StratPPI estimator, bootstrap-t limit
over-refusal (0.16)	100	0.023; 2.39	**0.066**; 4.11	0.042; 2.58
over-refusal (0.16)	200	0.024; 2.30	0.052; 3.46	0.042; 3.10
over-refusal, pushed by the Lagrangian (0.18)	100	0.024; 2.19	0.0558; 3.24	0.036; 2.43
over-refusal, pushed (0.18)	200	0.023; 2.13	0.045; 2.85	0.037; 2.49
non-refusal, encoded prompts (0.09)	100	0.034; 1.38	**0.086**; 2.59	0.045; 1.43
non-refusal, encoded prompts (0.09)	200	0.027; 1.36	**0.063**; 2.11	0.041; 1.59
refusal, harmful prompts (0.65)	100	0.047; 4.74	**0.0564**; 6.30	0.033; 1.77
refusal, harmful prompts (0.65)	200	0.040; 5.19	0.045; 6.29	0.034; 5.35
refusal, encoded prompts (0.93)	100	0.054; 0.96	0.032; 0.78	0.031; 0.83
refusal, encoded prompts (0.93)	200	0.045; 1.12	0.026; 0.92	0.033; 1.03

Reference-rate strata (section 8.1). In Table A1, bold marks a miss more than two Monte Carlo standard errors over delta, which is 0.0562 here; the two cells on either side of it carry a fourth digit. The predictor is constant in 5 to 7 of the 8 strata, and in those strata StratPPI is a stratified normal interval on the labels. At delta 0.1 StratPPI as published is over its level in 3 of 10 cells. At delta 0.05 it is also over in every one of the four rare-label cells, where `b1w` is over in none. The paper's own allocation rules do not repair the normal limit: 9 of 26 cells are over with the oracle rule and 11 with the heuristic. With a bootstrap-t limit the estimator is over its level in 6 of 26 cells under the oracle allocation and in 11 under the heuristic, against none of 28 under proportional allocation. Under proportional allocation it is ahead of `b1w` in seven cells, by 3–35%, behind by 8% and 14% in the two at a 93% rate, and far behind in one (1.77 against 4.74). A sweep places that drop at a safety set of 100 with 8 or 16 strata; with 4 strata, or with a safety set of 200 at any stratum count, it is absent. A stratified Wald-t limit on the same strata also held in all ten cells (largest miss 0.036, median ESS 2.06). In our predictor comparison the predictor is the same reference rate that defines the strata; splitting the reference samples, four for the strata and four for the predictor, did not change the picture (median ESS 1.97 against 2.10).

Judge-logit strata (section 8.2). StratPPI as published is over its level in 26 of 28 cells under each library's conventions, in 28 of 28 with the paper's oracle allocation and in 24 with its heuristic. The cleanest evidence is the judge wording with no constant stratum: 13 of 14 cells over, with misses from 0.056 (the one cell that is not over) to 0.187. With a bootstrap-t limit and the oracle allocation the estimator is over its level in 14 of 28 cells, against none under proportional allocation.

The 0.90 of Table 4 is the worst of several cells with one cause. It is the rubric judge at a 5% rate, 1,000 labels, 5 strata, heuristic allocation. The heuristic gives a stratum a share in proportion to its weight times the square root of the mean of $c(1-c)$ plus the variance of c, with c the judge's confidence. This judge's probability is under 0.001 in four of the five strata, so the rule sends 992 labels to the fifth and the four keep 2 each, a floor that is ours (the paper gives none). Those four strata hold 19% of the positives, at rates of 1.0–1.5%, and their eight labels show no positive in 91% of draws. The estimate is then about a fifth too low and the strata contribute nothing to its

estimated variance, so the limit misses in 90% of draws, under the normal limit and the bootstrap-t alike. The floor is not what fails: with 10 labels a stratum the same cell still misses in 59% of draws, and four other cells on this judge miss in 68% to 85%. Fisch et al. note, for one of their datasets, that the heuristic becomes too aggressive at larger n when the judge's confidence is not calibrated; this is that case, at a rare rate.

B The mechanism behind section 9.2

Our account of the mechanism, which no experiment here isolates, is a change in the form of the positives. The reward pulled toward long helpful answers and the multiplier penalised the guard's refusal flag. After training, the responses the guard flags are long refusals with an explanatory, helpful-sounding body: by the guard and a surface pattern (a refusal opener and more than 40 words), their share of the guard's flags rose from 37% in the reference to 82% in the trained policy. The rubric judge reads that body as an answer: of 93 guard-flagged refusals after training it called 90 answers. Human labels side with the guard. Of 18 such responses of the trained policy, the annotator labelled 12 refusals, 6 answers and none refuse-then-answer, and refuse-then-answer is 3.4% of the reference's responses against 0.3% of the trained policy's (section 10.3). We had first read these responses as a refusal sentence followed by the answer; the labels do not support that reading, and the surface pattern is not a label (on XSTest's completions 56% of responses with it are human full refusals and 34% partial ones).

C The design check of section 10.4

A check of the whole design on the guard's real scores, with synthetic labels drawn from the guard's logit at the rates of section 10.3's sheet in each band of the logit (a synthetic population with rates of 9.5% and 10.1%, a true difference of +0.006), puts the miss rates of the four limits at 0.001, 0.040, 0.045 and 0.042 for the pool rate against a level of 0.05, and their mean certified margins at 0.065, 0.037, 0.032 and 0.035. When the 490 prompts are redrawn each time, so that the claim is about their source, the four miss rates are 0.001, 0.051, 0.063 and 0.052 (Monte Carlo standard error 0.005); limit (b1) is not built for that case. The exact bound is about twice as wide as the approximate ones: with few discordant pairs in 300, exactness is the larger cost and the guard's gain is the smaller one. The synthetic labels disagree with the guard independently across the two policies, which weakens both the pairing and the guard, so these margins are on the wide side and those of Table 10 on the narrow side. The guard's gain in this check is about 1.4 in labels, against 2.5 in Table 10. In the guard's own terms, on the 3,920 guard-only responses per policy at 256 tokens, the refusal rate is 16.9% for the reference and 17.2% for the trained policy.

D What was fixed in advance, and where

Where this paper says a rule or an expectation was fixed before its result, the table gives the record. A commit that was made, and pushed to a remote, before the result existed is a record of the order that we could not have written afterwards. Where a note and its results entered the repository in one commit, the order rests on the note's own statement, and the table says so.

what was fixed	section	record
the first strata rule (ties kept together, cuts at quantiles, small strata merged) and the pre-flight's error criterion	8.1	a design note committed on 2026-09-27 ($9e1b7b4$), before the first results (1 afe 911, the same day). Neither was kept: the rule was replaced and the criterion failed. The truth used for coverage, fixed in the same note, was changed as well (section 8.1)
the expectations for the judge-assisted estimators (not the routing rule of Table 5, which came out of the results)	8.2, 9.1, 9.4	a note dated before the estimators were run, committed together with their results on 2026-09-30 ($3450d8c$): no independent timestamp
the reading of a failure at step 200	9.2	a note dated while the scoring pass ran and before its output, committed together with the result on 2026-10-01 ($3f89e98$): no independent timestamp
the hypotheses for the robot benchmark, against which the result on joint effort is marked exploratory	10.1	a note written after the termination counts and 12 transcripts had been seen, committed together with the results on 2026-10-02 ($50863a4$): no independent timestamp
the two-way bounds: definitions, fallbacks, seed, sizes and reading	10.2	a registration committed with its script on 2026-10-07 (cb 50927), before the script was run on the benchmark's tables. The commit reached the remote only with its results, 42 minutes later ($56721e6$), so the order rests on the local commit time
the confirmation pool: prompts, labels, class limits, checks, sizes, the GPU cap, eight predictions and their scoring code	8.1	a registration committed and pushed with the scripts and the prompts on 2026-10-08 (9 fb $817d$), before any response was generated
the mid-rate pool: the mixture, labels, checks, sizes, the GPU cap, ten predictions and their scoring code	8.1	a registration committed and pushed with the scripts, the prompts and the re-mix analysis on 2026-10-08 ($d56$ feb 6), before any response on its prompts was generated. The predictions were written after the same analysis of a re-mix of the first two pools' responses, which loosened one (the bootstrap-t StratPPI limit may be over in one cell, by at most 0.01), added one (`b1w` within 0.01 of its level at a rate between 0.45 and 0.55) and set the floor of a third (the Wald-t limit's gain, at 80% of the re-mix's smaller value); the registration names the first two as changes and gives the third's origin. Two more differences from the first registration were settled with the re-mix in view: the prediction on the cap is for the Wald-t limit's gain and not `b1w`'s, and `b1w` with the sampled-pool term carries no prediction
the analysis of the human-label certificate	10.4	the rules and the analysis script, committed on 2026-10-04 ($9e45371$, $e076974$) before any label of this sample. They replaced a first version of the same day ($f1582e5$) after the labels of section 10.3 had shown a difference near zero. The treatment of a zero-variance resample in the approximate limits was amended once, on 2026-10-07 ($a5$ bd $04b$), also before any label of this sample, with the design check unchanged. No label has been collected

Table D1. The eighteen registered predictions of section 8.1 and their outcomes. "Large pool" is the redraw with replacement at 40,000 draws; "source" is the claim about the prompt source at 10,000 replications; a cell is one safety-set size (100 or 200) at one delta (0.05 or 0.10); gains are at delta 0.05 against the pooled Wilson bound on a random draw. A label is mid-rate between 5%

and 95%.

	prediction, as registered	**outcome**
	*First registration (9 fb 817*d*): three pools, refusal rates 12%, 82% and 98%*	
P1	large pool: the Wald-t limit and the bootstrap-t StratPPI limit are over in no mid-rate cell	refuted: the bootstrap-t limit is over in 1 of 8 cells (0.105 at delta 0.10, at 82%); the Wald-t limit in 0 of 8
P2a	large pool: `b1w` is over in no cell at a rate under 0.45	kept: 0 of 4 cells at 12% (0 of 12 with the two rare labels)
P2b	large pool: `b1w` is over in at least one of its four cells on every mid-rate label above 0.55	kept: 4 of 4 at 82%
P3	the gain of `b1w` with a large pool is within 20% of the pre-flight's prediction on every mid-rate label	kept: 1.53 and 1.62 against 1.82 at 12%; 1.49 and 1.62 against 1.65 at 82%
P4a	source: `b1w` with the sampled-pool term is over in no mid-rate cell at delta 0.05, and the Wald-t limit with the term in none at either delta	refuted: `b1w` is over in 1 of 4 cells (0.058 at 82%, safety set 100); the Wald-t limit in 0 of 8
P4b	source: `b1w` without the term and the bootstrap-t StratPPI limit are each over in at least a third of the mid-rate cells at delta 0.05	kept: 3 of 4 and 4 of 4
P4c	source: the gain of `b1w` with the term is within 15% of the cap on every mid-rate label	kept: 1.32 and 1.22 against 1.35 and 1.24 at 12%; 1.22 and 1.20 against 1.33 and 1.24 at 82%
P5	a rare label: the strata gain nothing (ESS of `b1w` between 0.9 and 1.15) and Clopper-Pearson on a random draw is over in no cell	kept: 1.01 to 1.11; 0 of 8 cells
	*Second registration (*d*56 feb 6), written after the re-mix: one mixed pool, refusal rate 46%*	
M1	large pool: the Wald-t limit is over in no cell	kept: 0 of 4 (misses 0.026 and 0.036 at delta 0.05, 0.063 and 0.083 at 0.10)
M2	large pool: the bootstrap-t StratPPI limit is over in at most one of four cells, and above its level by at most 0.01 in any	kept: 0 over, 3 unresolved, largest excess 0.001
M3a	large pool: `b1w` is over in no cell at a rate under 0.45, and in at least one of four above 0.55	not tested: the rate, 0.459, lies between
M3b	at a rate between 0.45 and 0.55, the miss of `b1w` is within 0.01 of its level in every cell	kept: 0.044 and 0.045 at delta 0.05, 0.096 at 0.10
M4	the gain of `b1w` with a large pool is within 20% of the pre-flight's prediction	kept: 2.78 and 2.97 against 2.79
M5	the gain of the Wald-t limit with a large pool is at least 1.7	kept: 2.09 and 2.56
M6	source: the Wald-t limit with the term is over in no cell	kept: 0 of 4
M7	source: `b1w` without the term and the bootstrap-t StratPPI limit are each over in at least one of the two cells at delta 0.05	kept: 2 of 2 each (misses 0.079 to 0.120)
M8	source: the gain of the Wald-t limit with the term is above 1 and within 15% of the cap	kept: 1.49 and 1.34 against 1.64 and 1.44
M9	a rare label: ESS of `b1w` between 0.9 and 1.15, and Clopper-Pearson over in no cell	kept: 1.00 and 1.08; 0 of 4 cells

E Reproducing the numbers

The numbers in this paper come from the scripts below and from the scripts of three earlier measurements kept with their results (the carried calibration and the sheets of sections 9.1, 9.2, 9.4 and 9.5; the robot benchmark's counts of section 10.1; the design effects of section 9.3), apart from those cited from the companion paper and a few recomputed from the data files. The scripts that

sample or judge responses need a GPU. The working draft carries a source tag on each number and a table that resolves each tag to a file.

what	script	output
Table 4, every validity cell under one rule; the table and the counts in the text asserted against the files	scripts/validity_recount.py	results/paper/validity_recount.md
section 7.1, the reference-rate-strata cells redrawn with replacement	scripts/replacement_check.py	results/paper/replacement_check.md
section 7.1, the exact miss probability of the Wilson limit	scripts/wilson_exact.py	results/paper/wilson_exact.md
Table 4, the labels-alone rows by enumeration	scripts/binomial_rows.py	results/paper/binomial_rows.md
Table 4, the four judge-plasmode cells with 20,000 unlabelled responses, at 4,000 draws	scripts/plasmode017_big.py	results/paper/plasmode017_big.json
section 8.1, the strata for a claim about the prompt source	scripts/twophase_check.py	results/paper/twophase_check.md
section 8.1, the confirmation on new prompts (registered)	scripts/confirm_pool.py	results/paper/confirm/confirm.md
section 8.1, the mid-rate pool (registered)	scripts/confirm_mid.py	results/paper/confirm/mid/confirm.md
end-point tests of every bound used	tests/test_bound_endpoints.py	run with pytest
sections 8.1 and 8.2, the StratPPI comparison	scripts/stratppi_baseline.py	results/paper/stratppi.md
sections 8.1 and 8.2, implementation check, allocations, PPBoot, sweeps	scripts/stratppi_validate.py	results/paper/stratppi_validate.md
section 9.2, the judge against human labels	scripts/judge_on_labels.py	results/labels/refusal/rubric_vs_human.md
Appendix A, the heuristic allocation in its worst cell	scripts/stratppi_heur_cell.py	results/paper/stratppi_heur_cell.md
section 10.1, the robot benchmark's sampling unit	scripts/robodojo_sampling.py	results/paper/robodojo_sampling.md
section 10.2, AgentDojo	scripts/agentdojo_recheck.py	results/paper/agentdojo_recheck.md
section 10.2, the two-way bounds (registered)	scripts/agentdojo_twoway.py	results/paper/agentdojo_twoway.md
section 10.3, the guard on XSTest	scripts/xstest_guard.py	results/labels/xstest/analysis.md
section 10.3, the refusal sheet	scripts/refusal_sheet_build.py, scripts/refusal_labels.py	results/labels/refusal/analysis.md
section 10.4, the label budget	scripts/p9_budget.py	results/paper/p9_budget.md
section 10.4, the prepared certificate and its design check	scripts/p9_sample.py, scripts/p9_certificate.py	results/labels/p9/design_check.md
figures	scripts/paper_figures.py	reports/figs/
references checked against arXiv and CrossRef	scripts/bib_check.py	printed report

Results cited from the companion paper on training (parts of section 7.2) are printed rates whose per-draw data were lost; they cannot be regenerated without retraining. The rows of Table 4 on a trained-policy harm pool are the exception: they are reproduced by enumeration.